

Quintile Return Book

Average monthly return of each factor's quintile portfolios and the long / short quintiles selected from them. 272 factors, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

Long Short Neutral

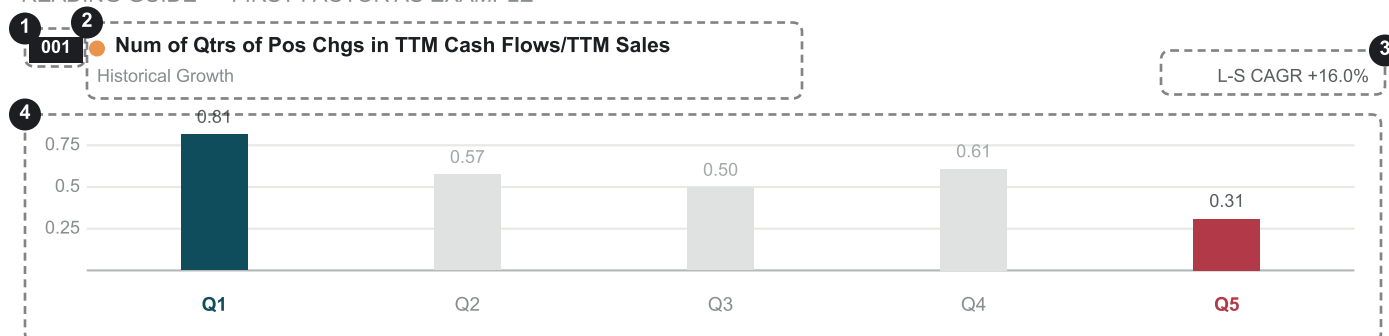
Long / short quintiles follow the live 48-month rules — spread-threshold based, not fixed to Q1 / Q5. Grey quintiles are left out of the portfolio.

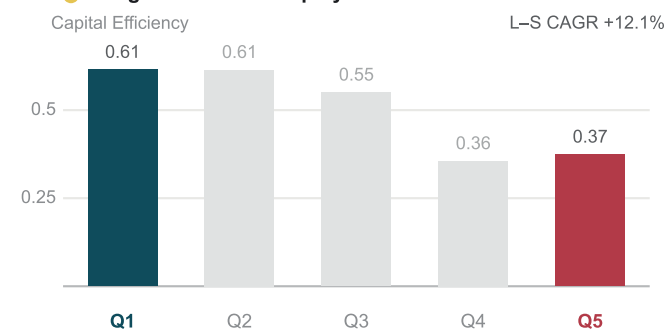
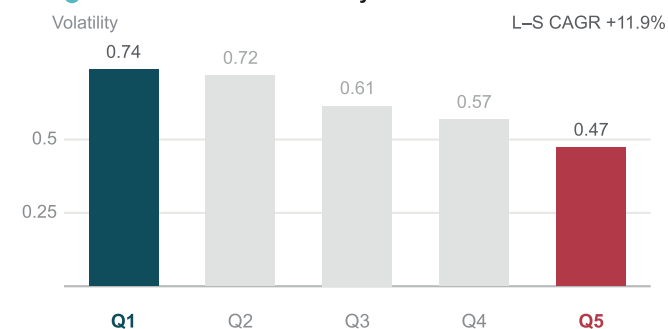
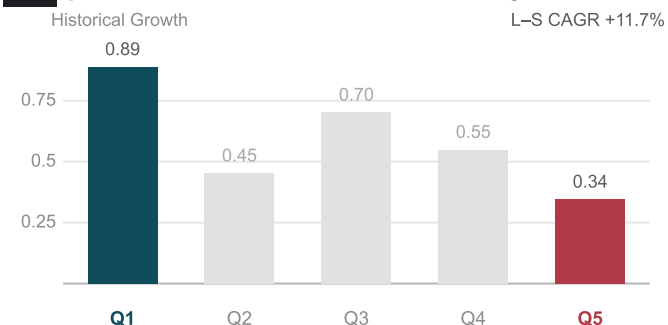
FACTOR STYLES

Valuation	42	Price Momentum	54
Earnings Quality	55	Size	3
Analyst Expectations	24	Historical Growth	56
Capital Efficiency	28	Volatility	10

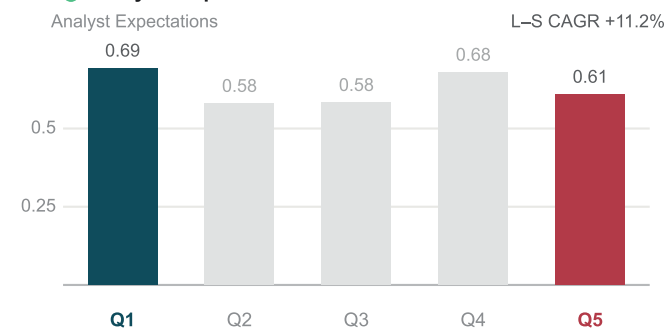
001 In current model portfolio (50 factors)

READING GUIDE — FIRST FACTOR AS EXAMPLE

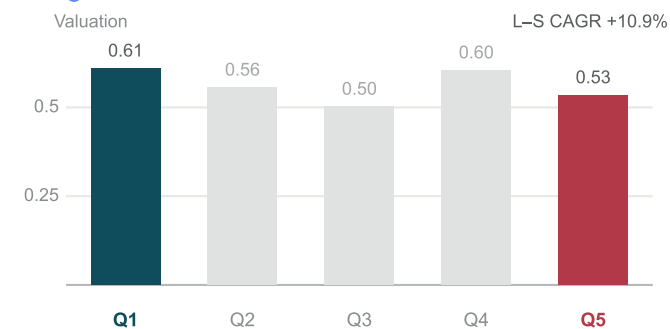


001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-M...****003 ● Long Term Debt to Equity Ratio****004 ● 12M Realized Price Volatility****005 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flo****006 ● Change of Asset Turnover**

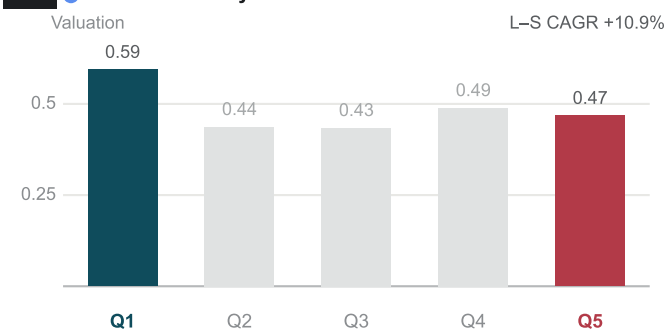
007 ● Analyst Dispersion for FY2 EPS



008 ● Assets to Price Ratio



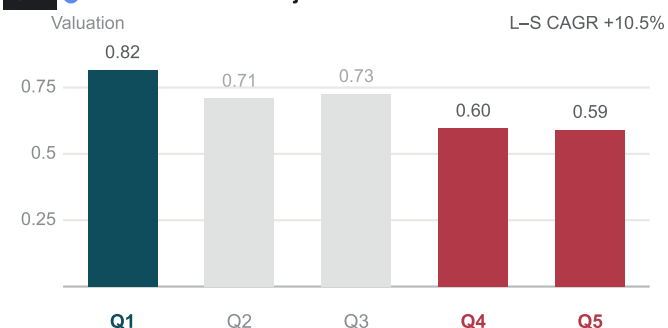
009 ● Current Liability to Price



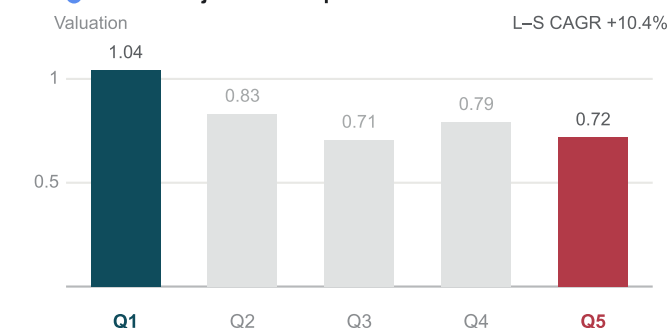
010 ● Number of Consecutive Qtrs of Positive Changes in TTM Quart...

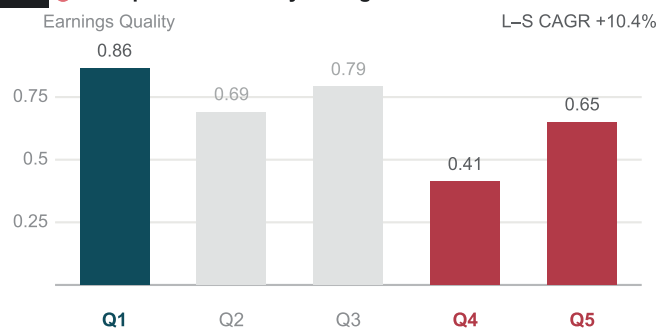
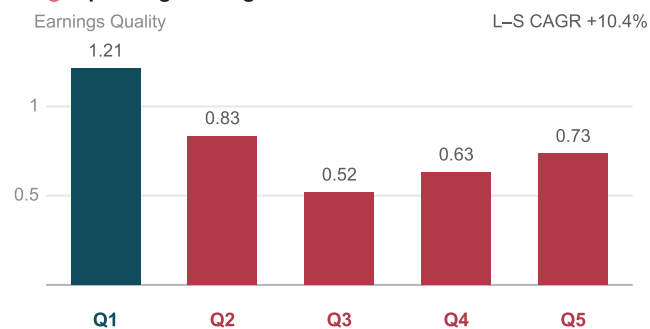
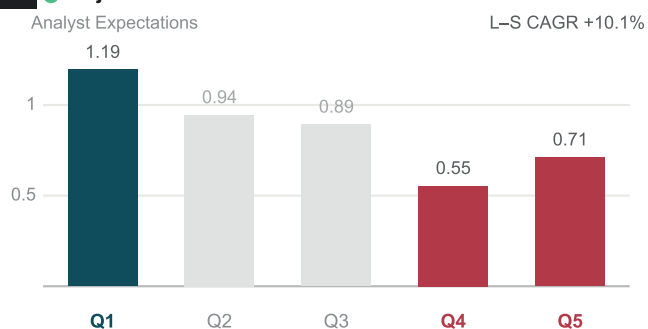
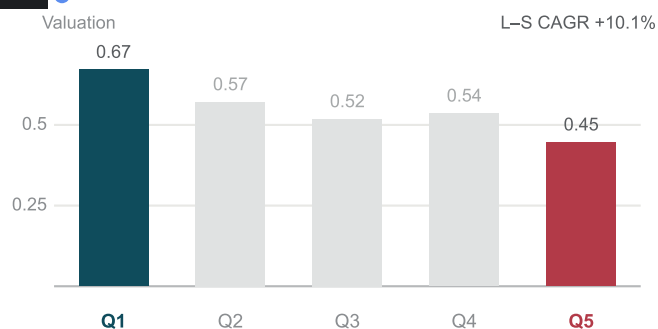
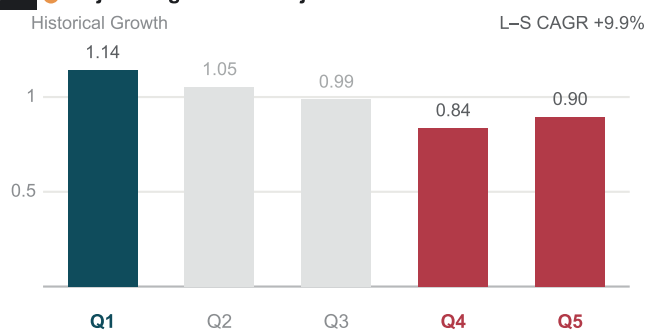
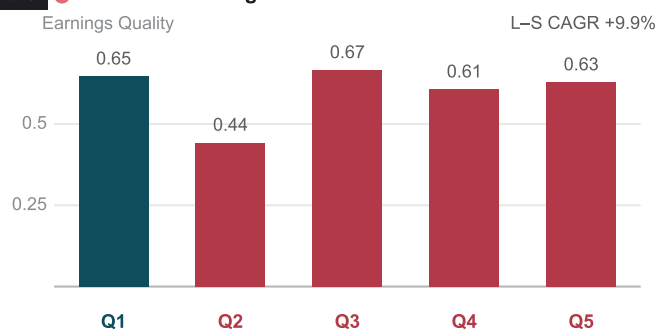


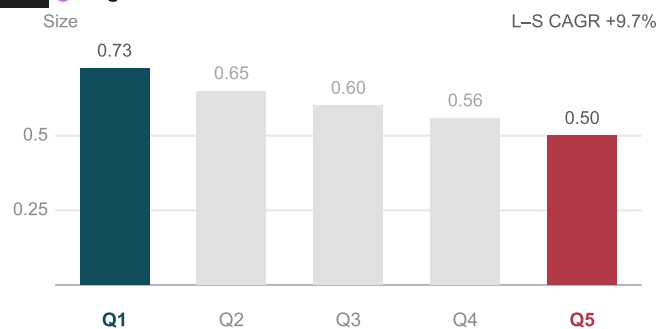
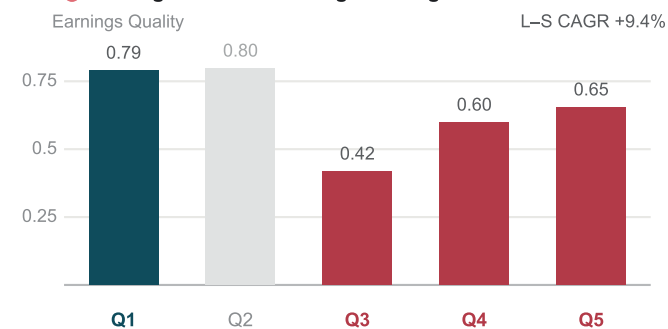
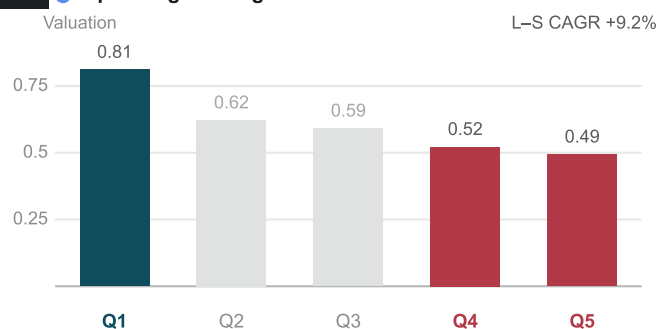
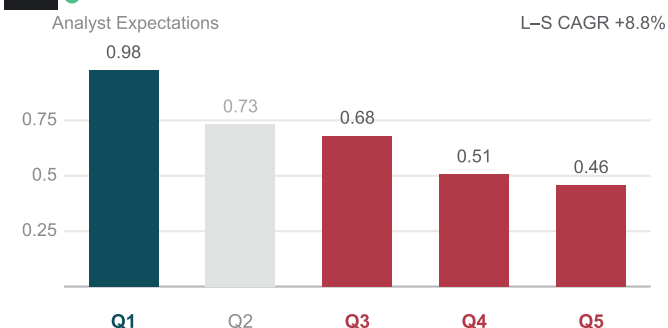
011 ● Inverse P/E Ratio Adj for Growth and Yield

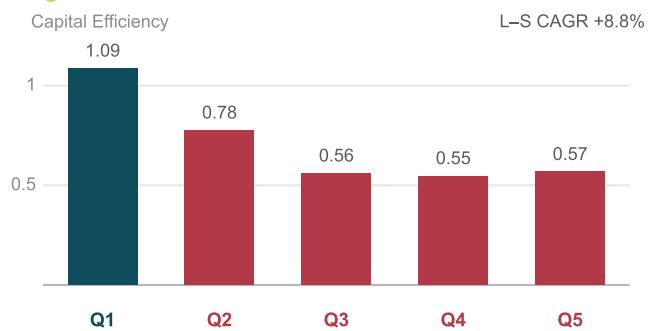
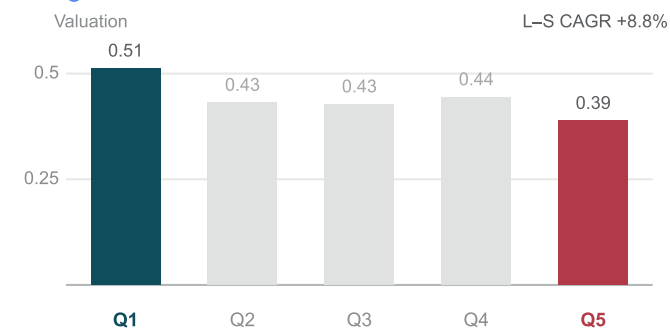
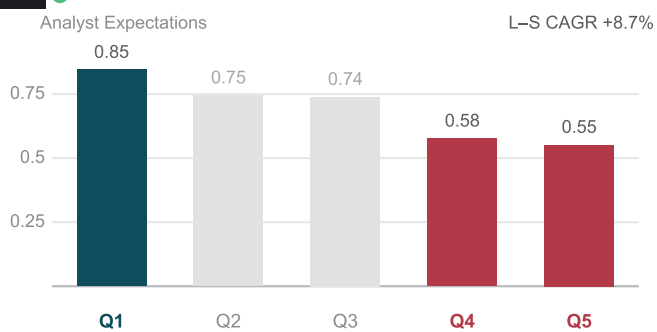
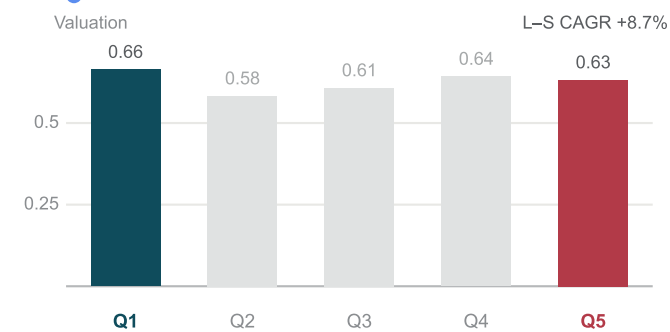
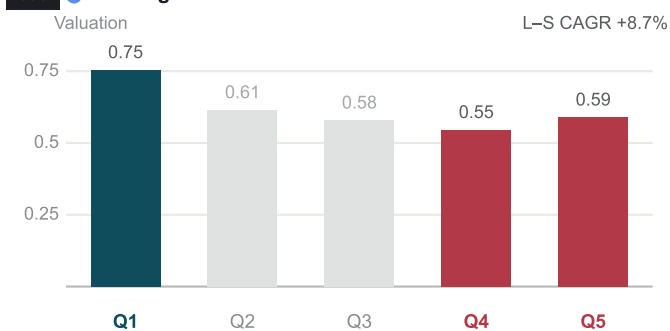


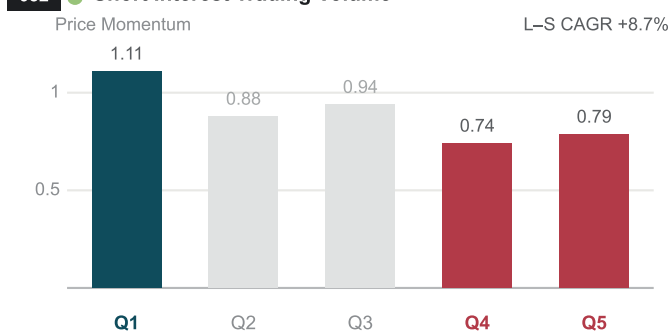
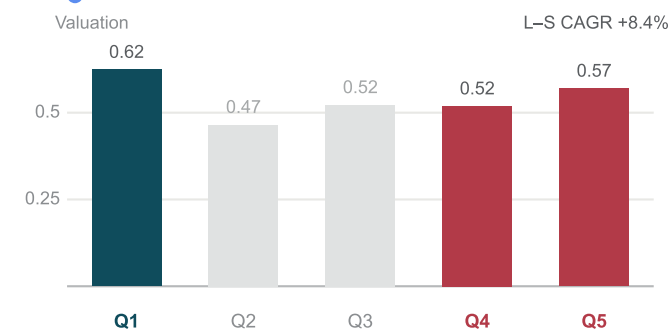
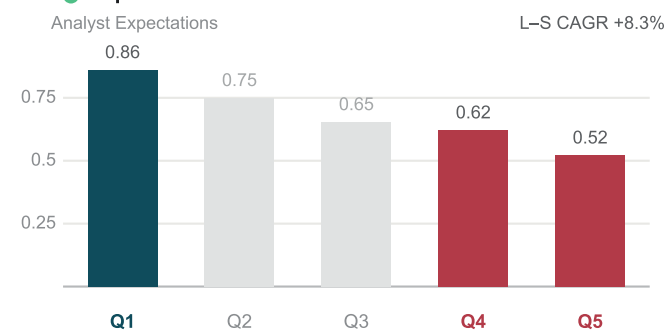
012 ● Cash to Adjusted Enterprise Value

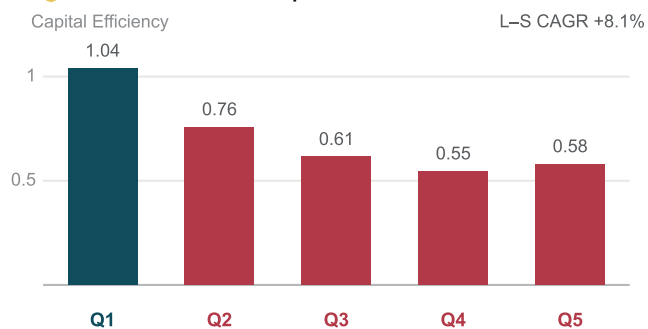
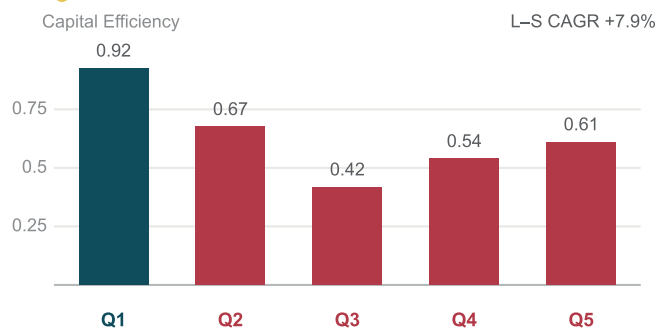
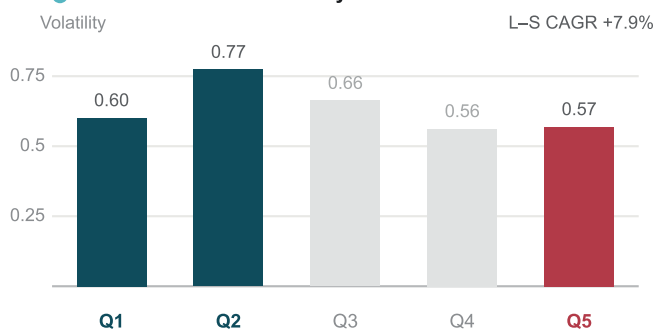
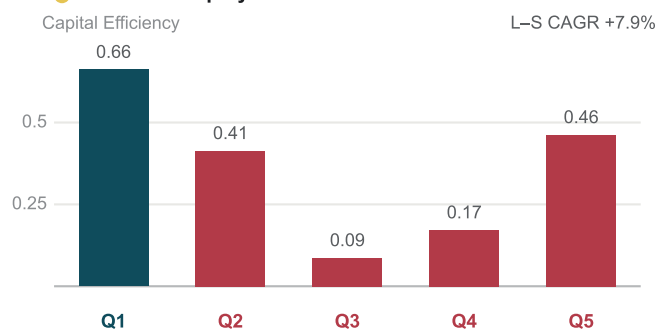
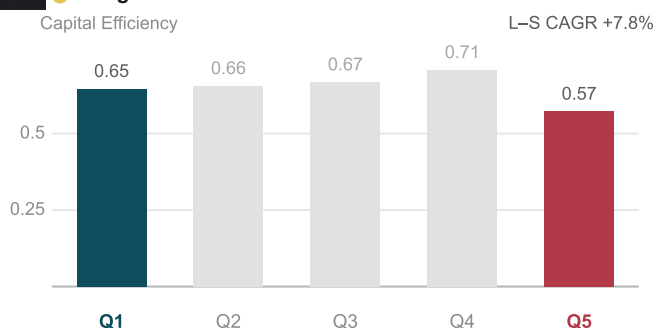
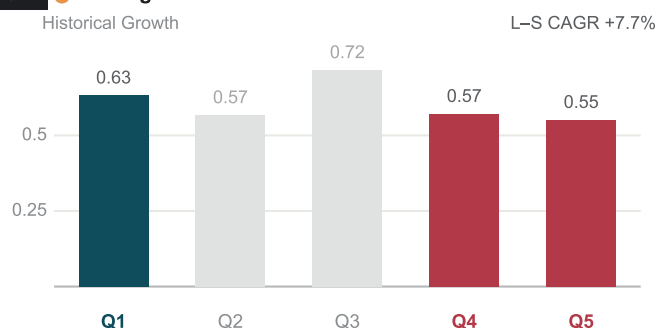


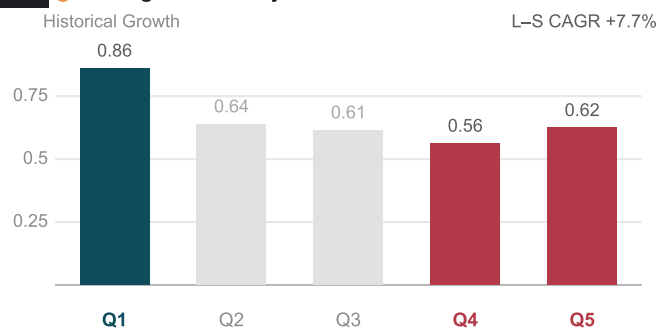
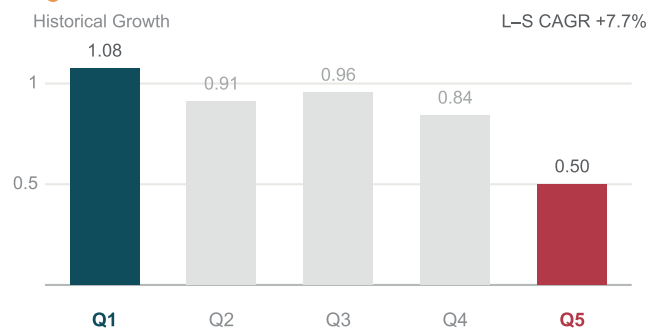
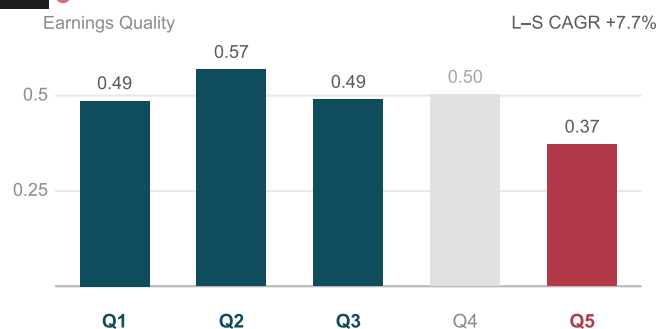
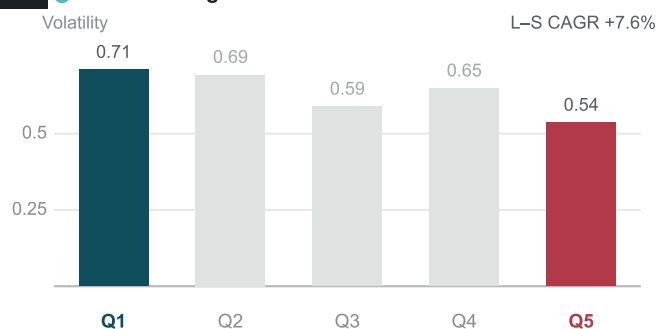
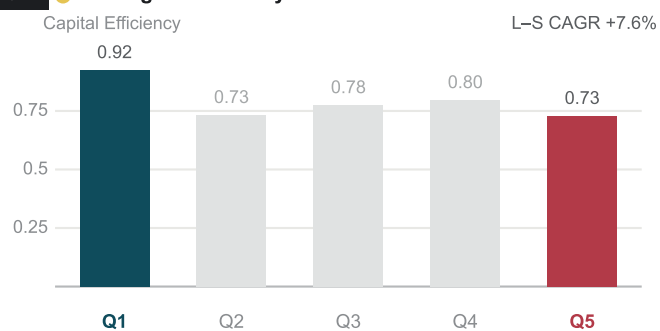
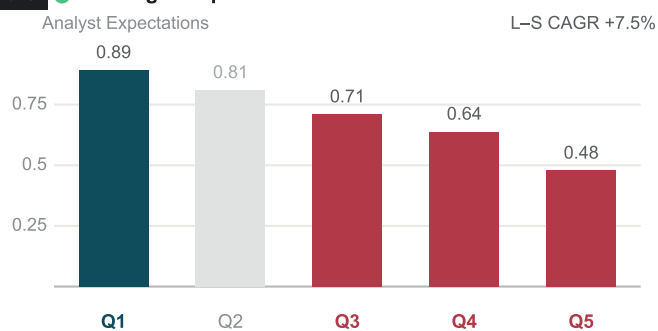
013 ● Unexpected Inventory Change**014 ● Operating Earnings to Assets Ratio****015 ● Adjusted 3M Revision in FY1 EPS Estimate****016 ● Sales to Price Ratio****017 ● Adj 1Y Chg in Asset Adjusted FCF****018 ● Gross Profit Margin**

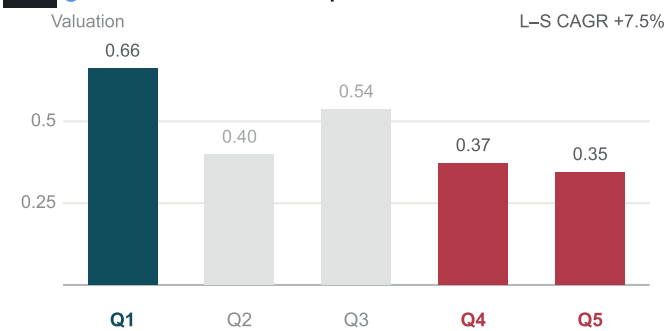
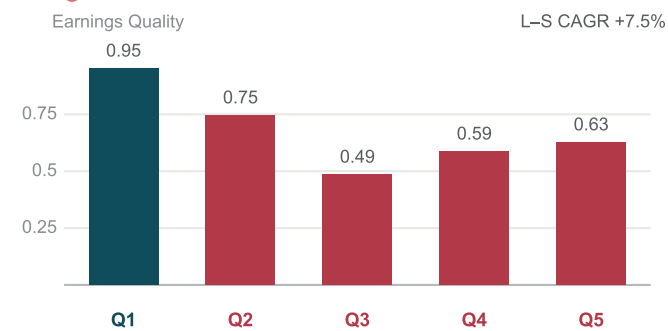
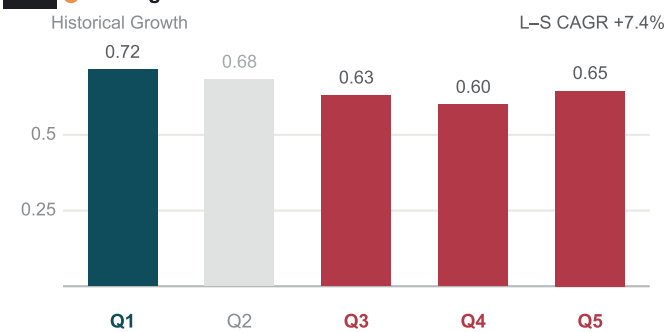
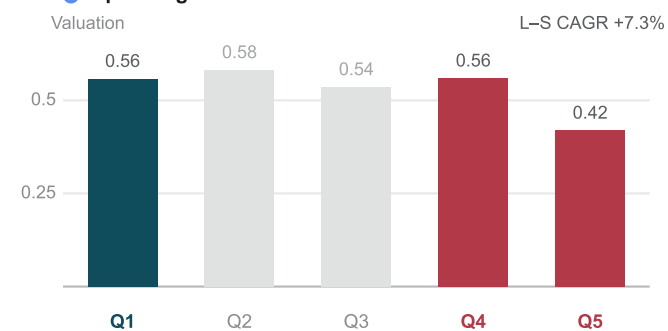
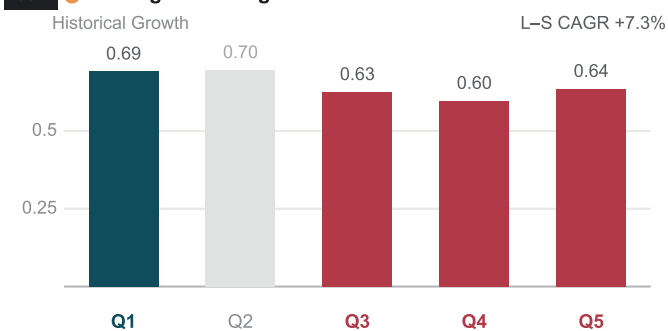
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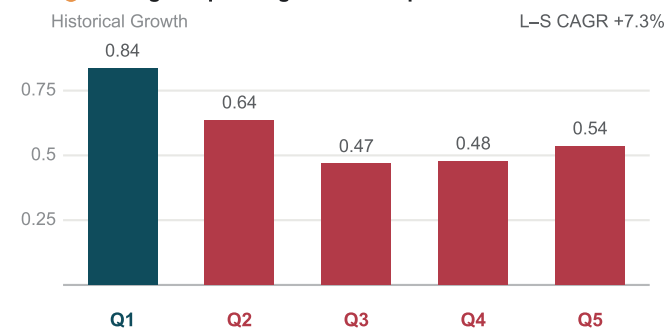
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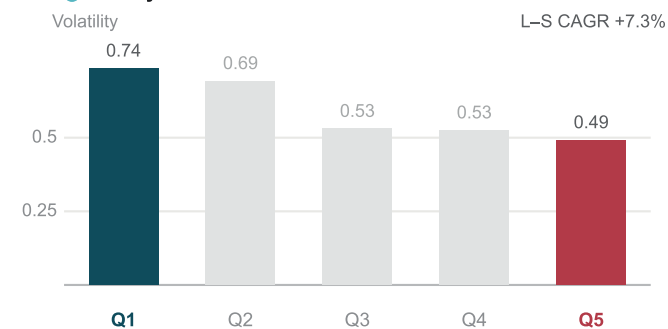
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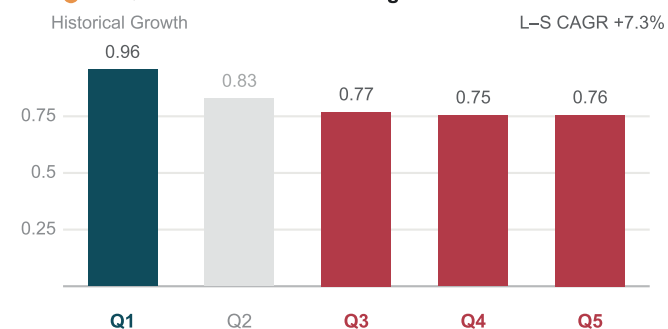
055 ● 3Y Chg in Operating Cash Flow per Share



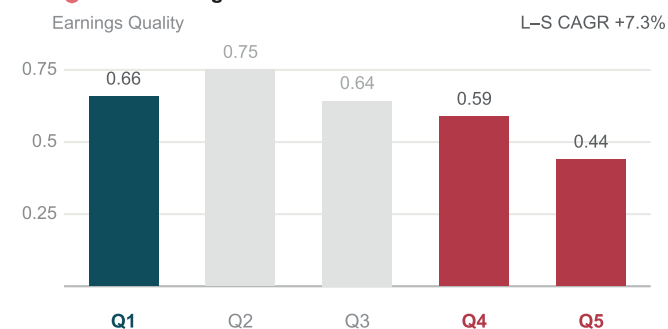
056 ● 90 Day Coefficient of Variation



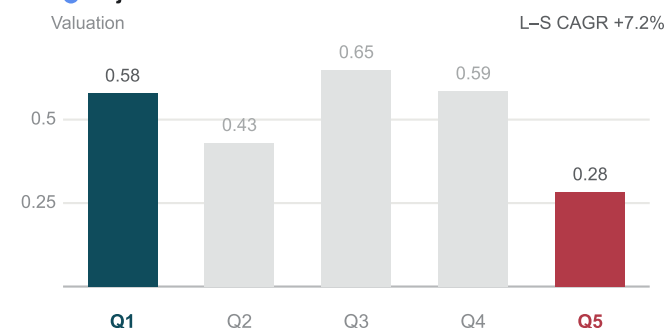
057 ● 12-Quarter Trendline in Trailing 12-Month Cash Flows



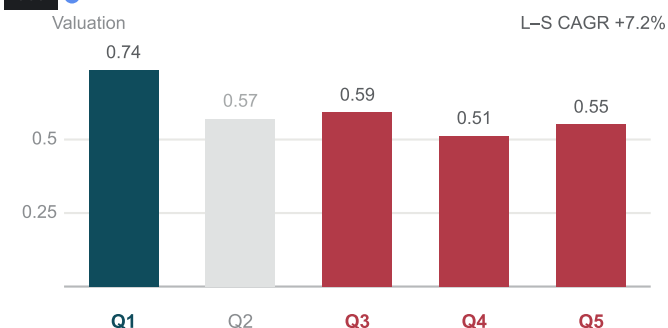
058 ● EBITDA Margin



059 ● Adj Free Cash Flow to Price



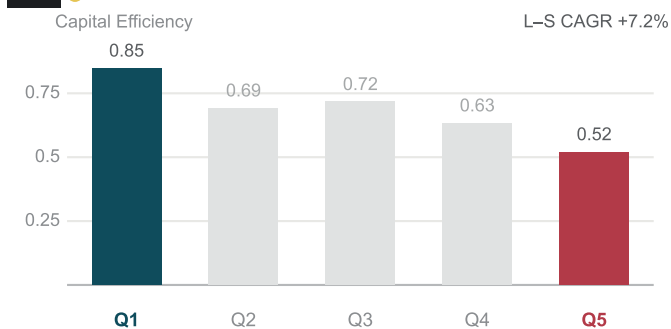
060 ● Normalized EP



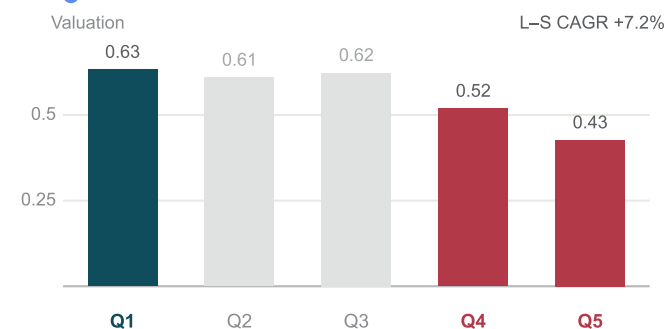
061 ● Working Capital to Sales



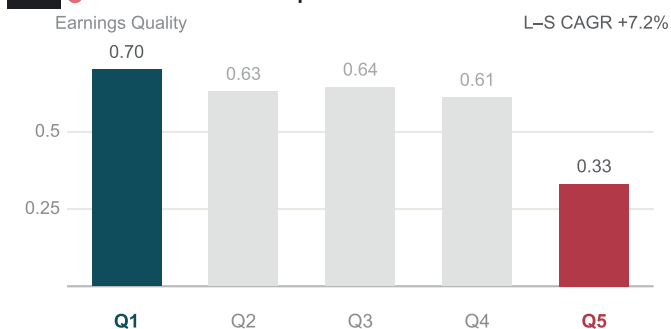
062 ● Receivables Turnover Ratio



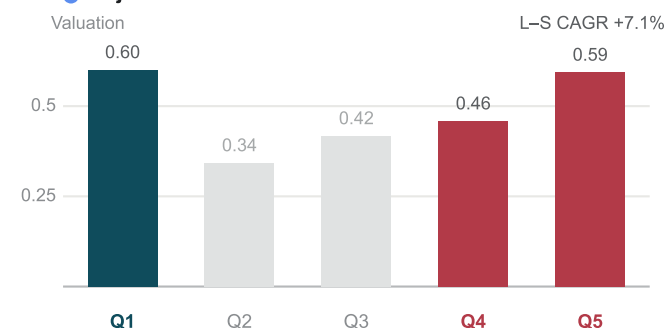
063 ● Dividends to Price Ratio



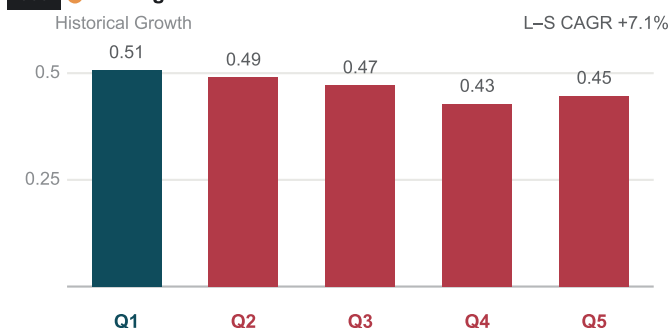
064 ● Sales to Invested Capital

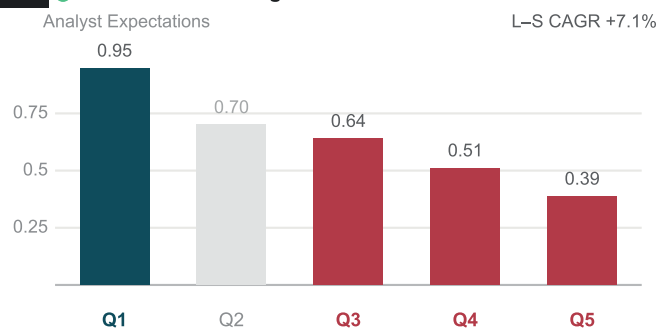
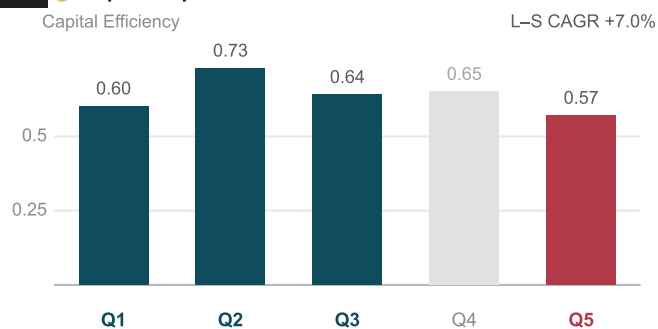
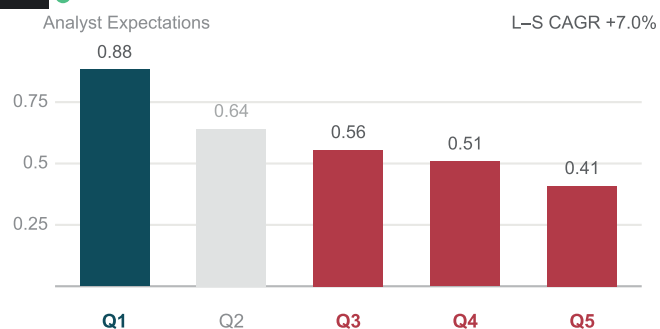
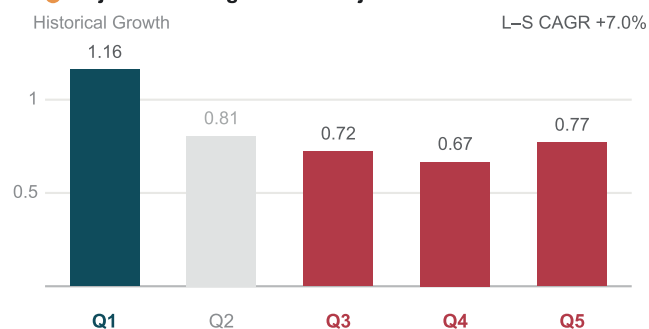
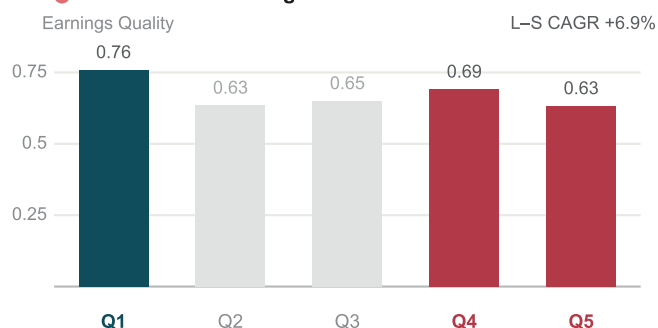


065 ● Adjusted EBIT to Price

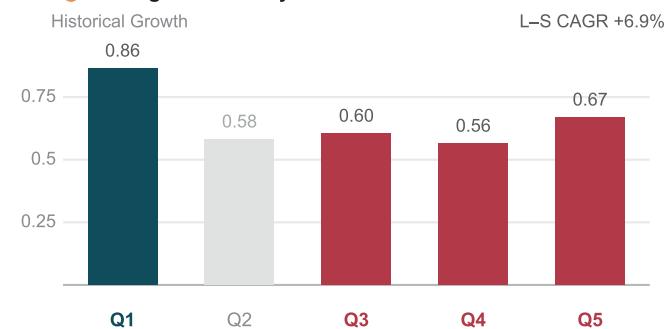


066 ● 3Y Chg in Cash Flow to Price

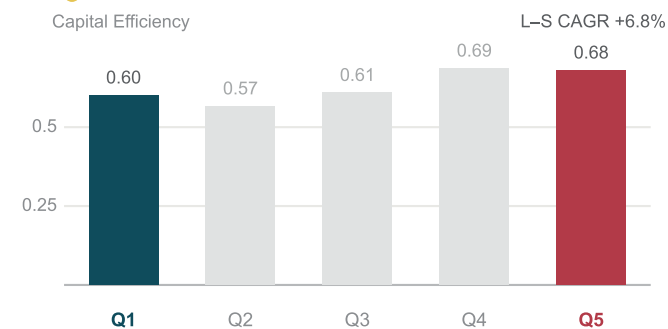


067 ● Street Revision Magnitude**068 ● Capital Expenditure to Total Assets****069 ● 3M Revision in FY2 EPS Estimate****070 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow****071 ● Analyst Dispersion for FY1 EPS****072 ● Year over Year Change in SGA to Sales**

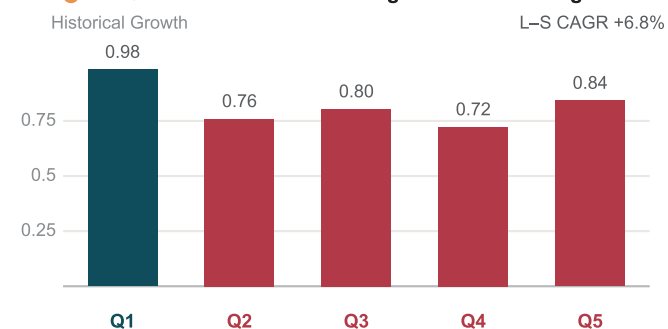
073 ● 1Y Chg in Asset Adjusted Cash Flow



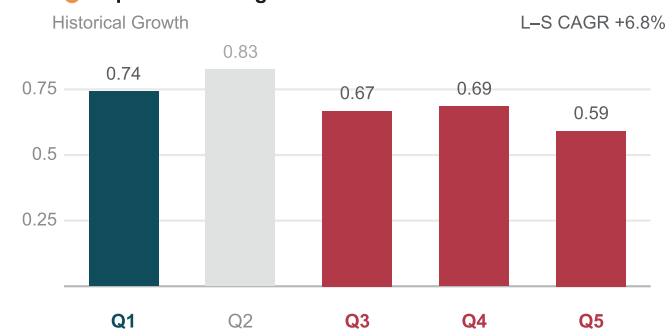
074 ● Tobin's Q Ratio



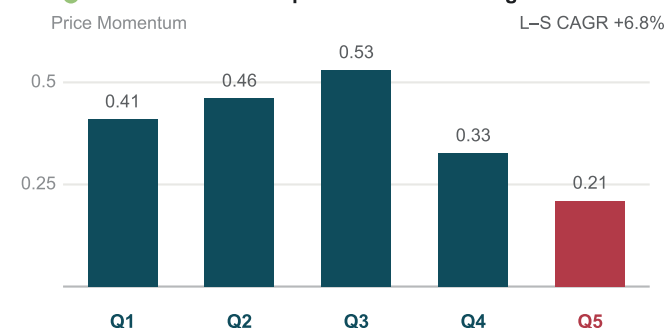
075 ● 12-Quarter Trendline in Trailing 12-Month Earnings



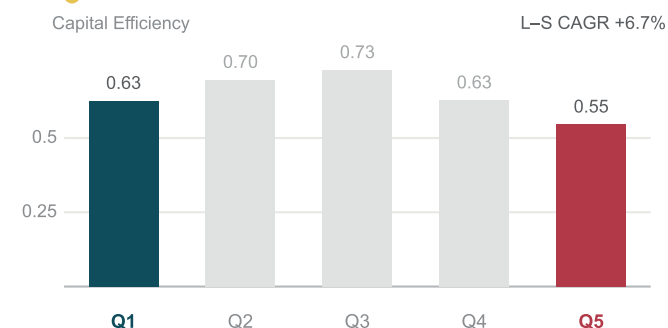
076 ● Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1

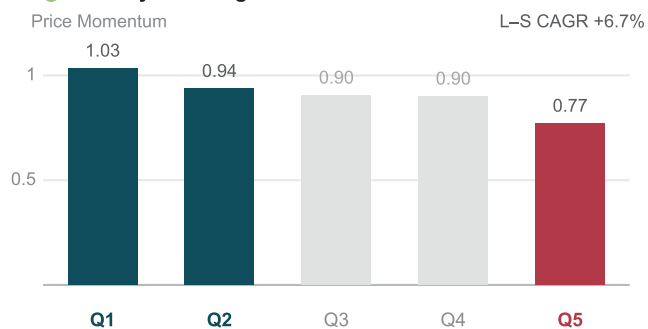
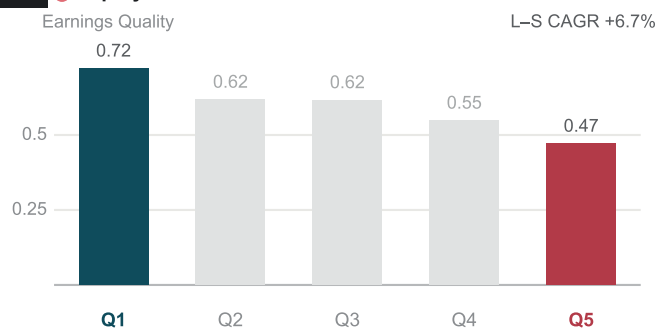
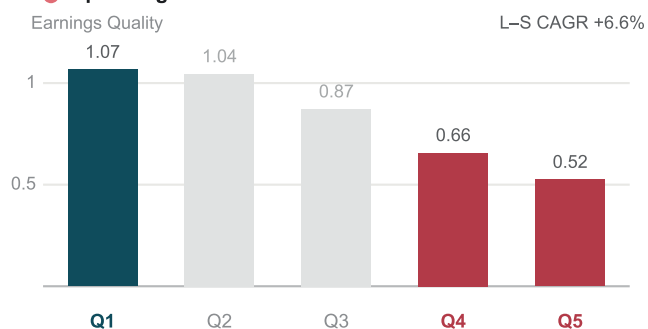
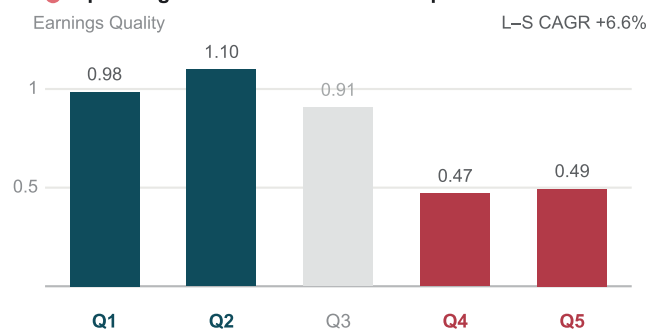
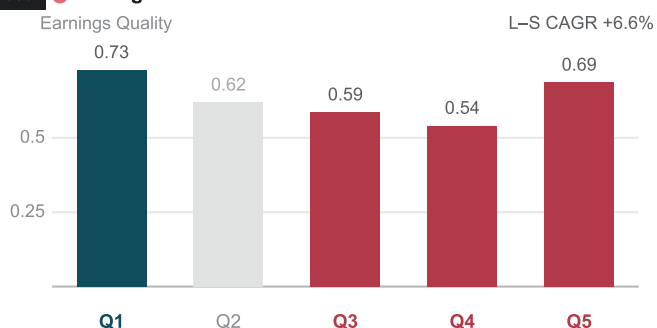
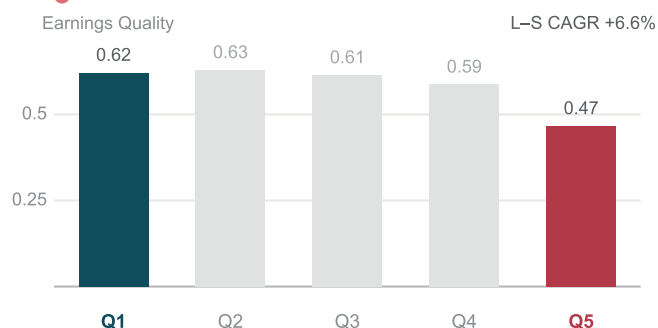


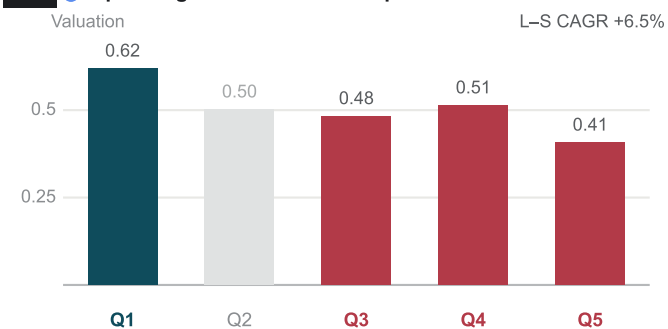
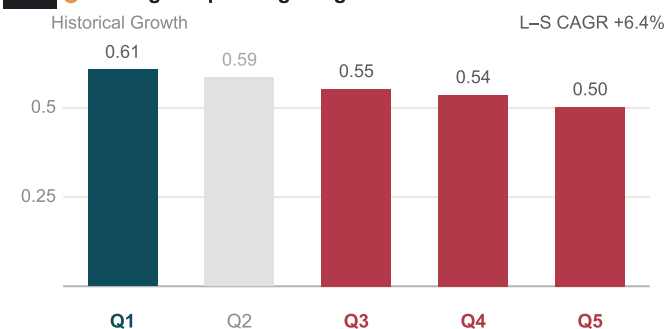
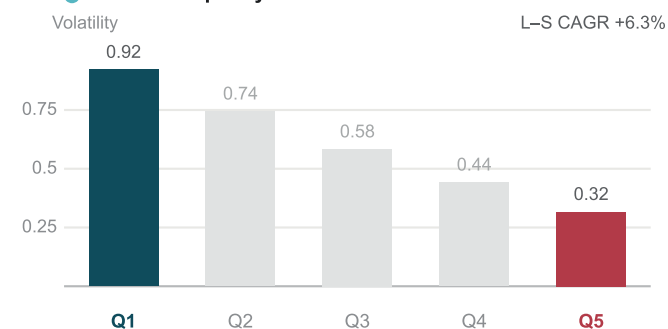
077 ● Institution Ownership Level - Active Manager



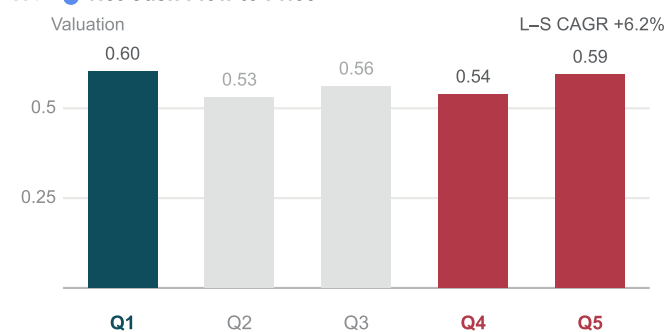
078 ● Debt to Assets Ratio



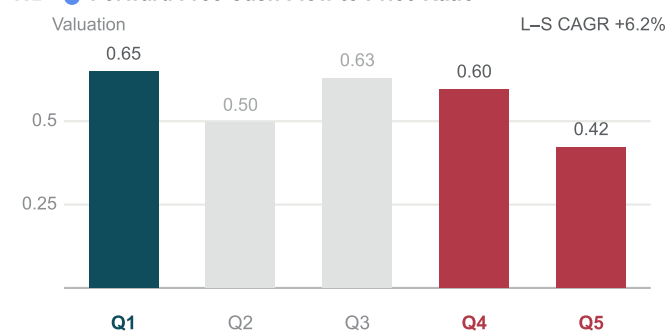
079 ● **Stability of Change in Number of Holders**080 ● **Equity Turnover**081 ● **Operating Cash Flow to Asset**082 ● **Operating Cash Flow to Invested Capital**083 ● **Change EPS to Sales**084 ● **Balance Sheet Accruals**

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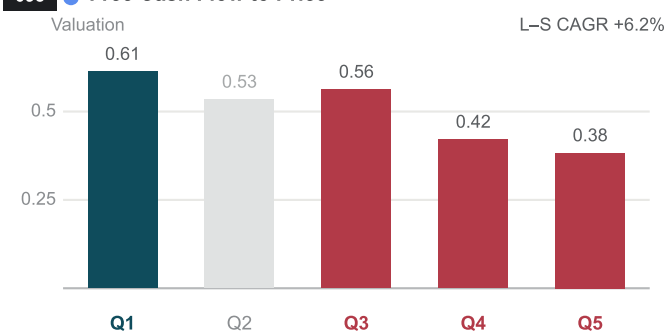
091 ● Net Cash Flow to Price



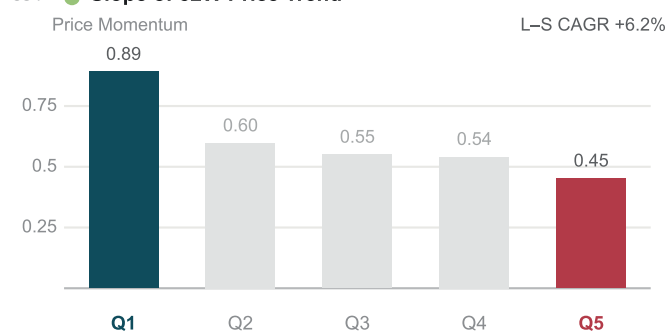
092 ● Forward Free Cash Flow to Price Ratio



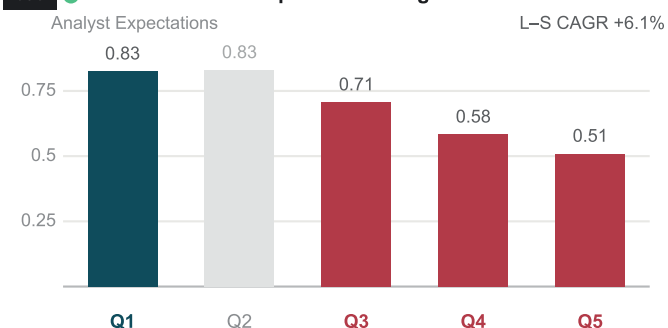
093 ● Free Cash Flow to Price



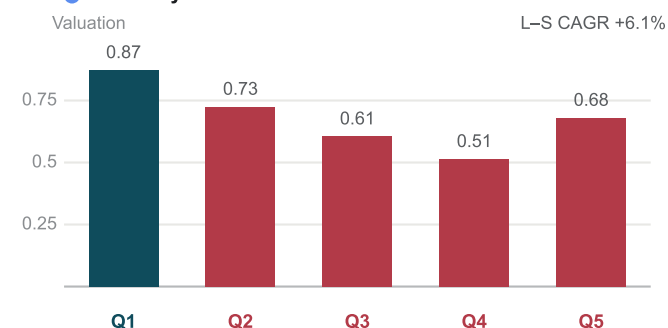
094 ● Slope of 52W Price Trend



095 ● Standardized Unexpected Earnings



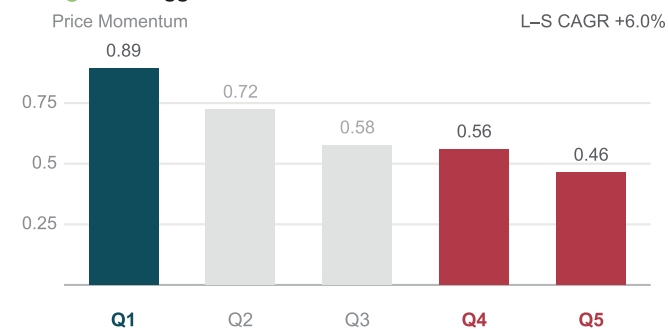
096 ● Solvency Ratio



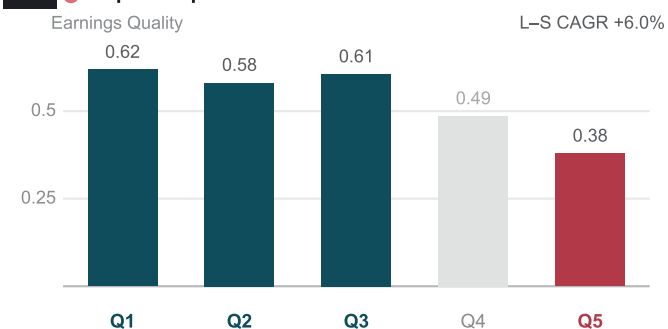
097 ● Inventory Turnover



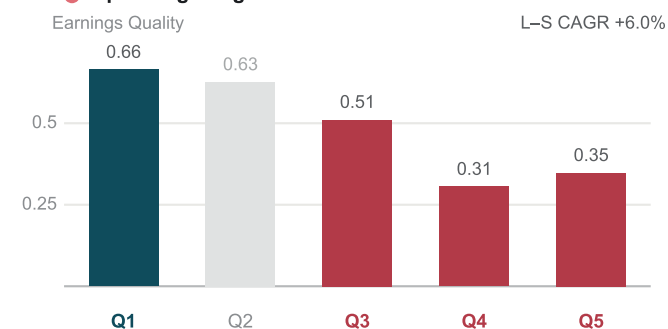
098 ● 39W Lagged Return



099 ● Capital Expenditure to Sales



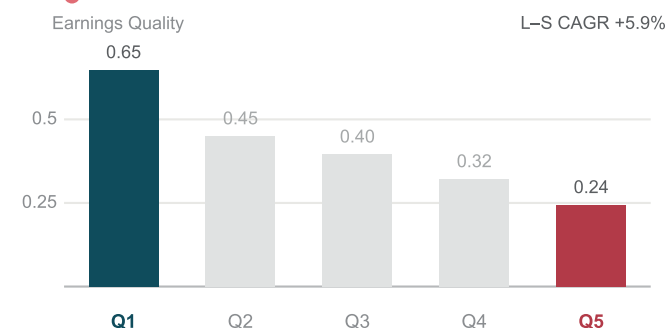
100 ● Operating Margin

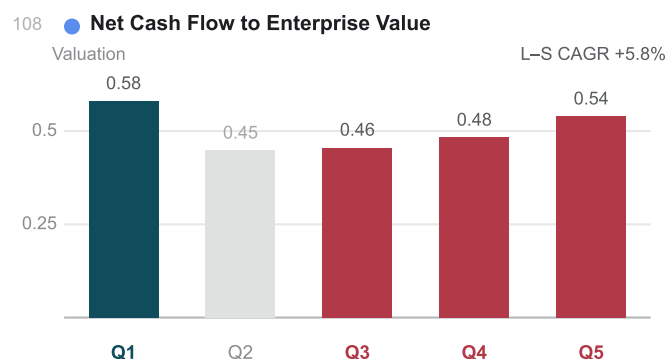
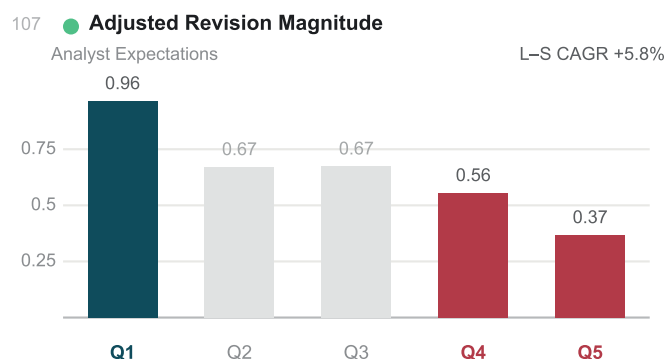
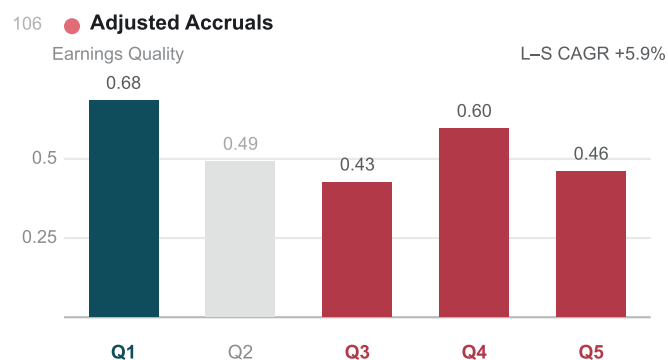
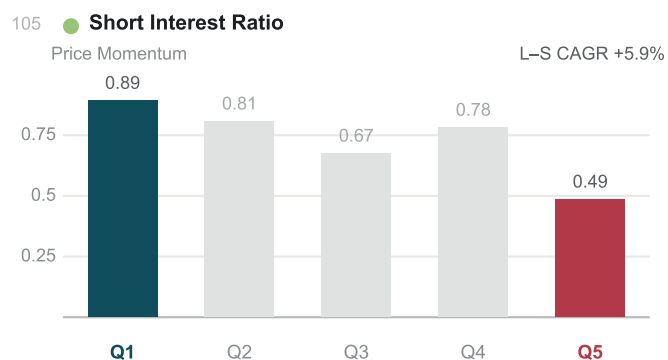
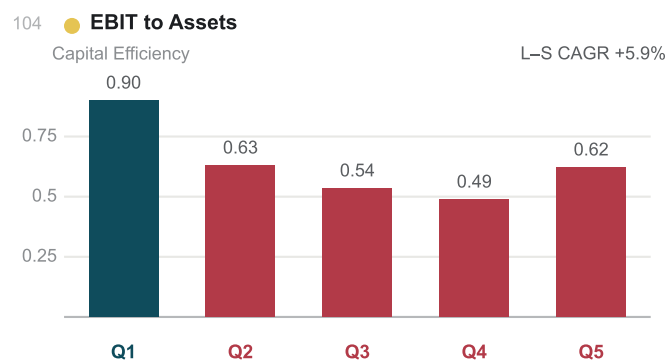
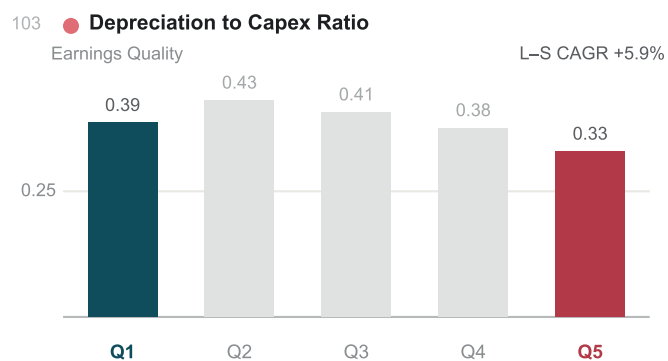


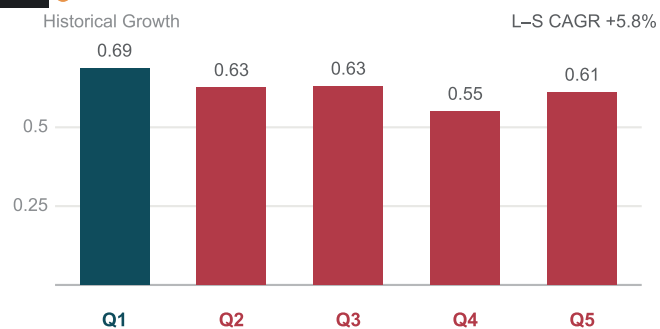
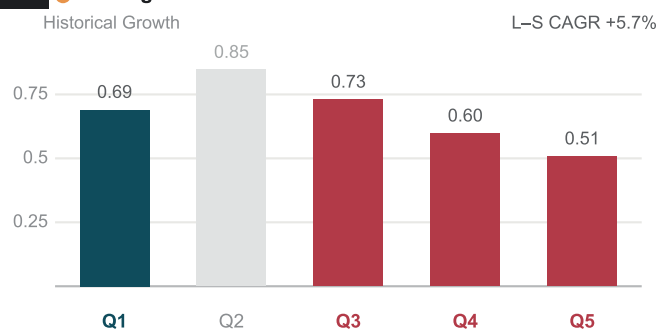
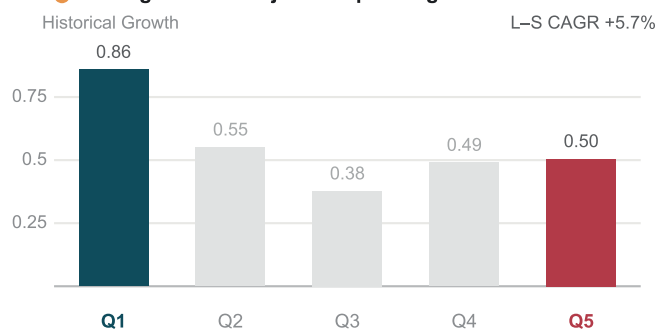
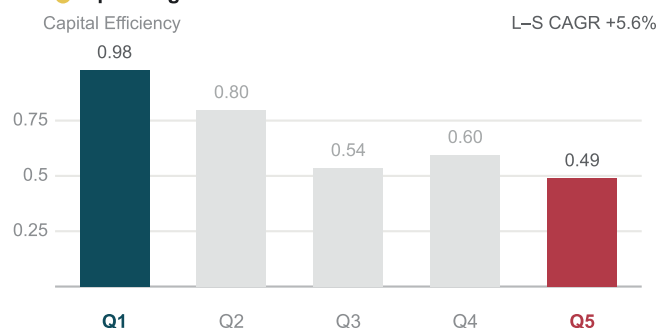
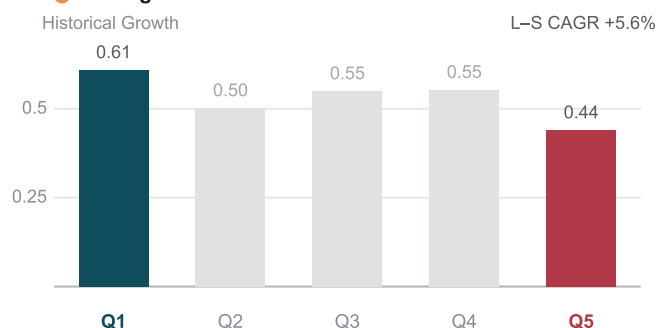
101 ● 1Y Chg in Operating Cash Flow per Share

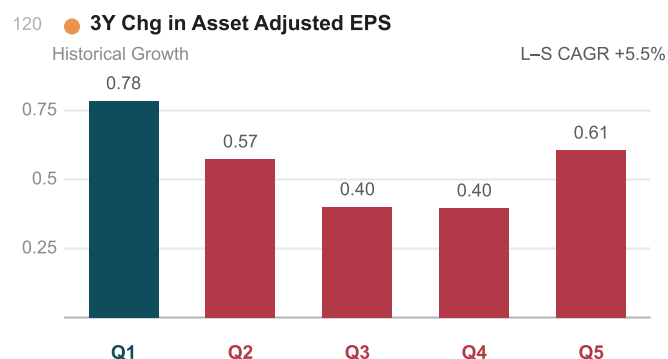
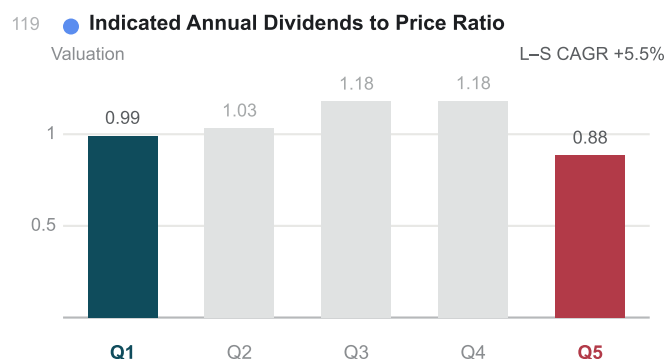
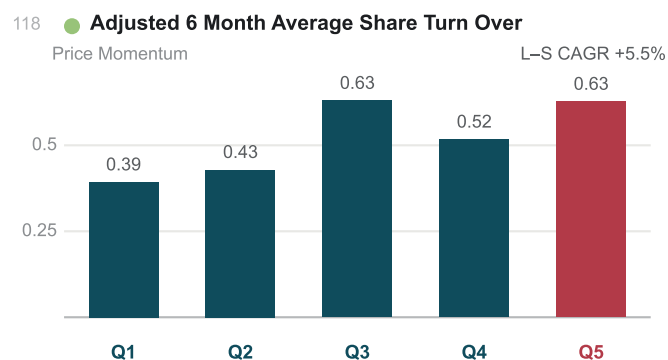
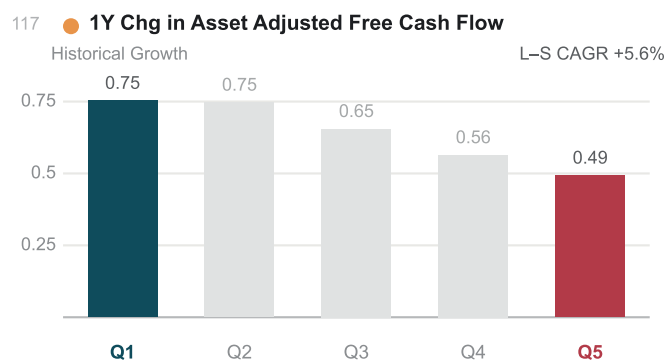
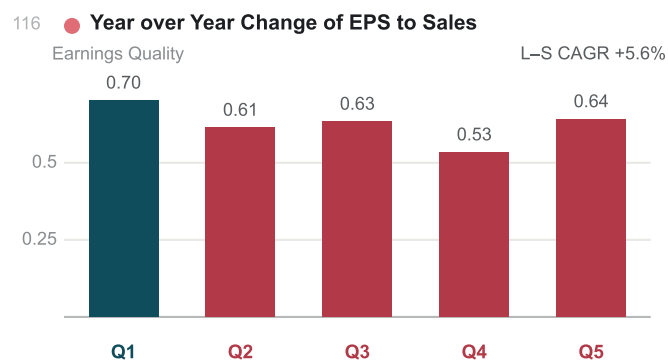
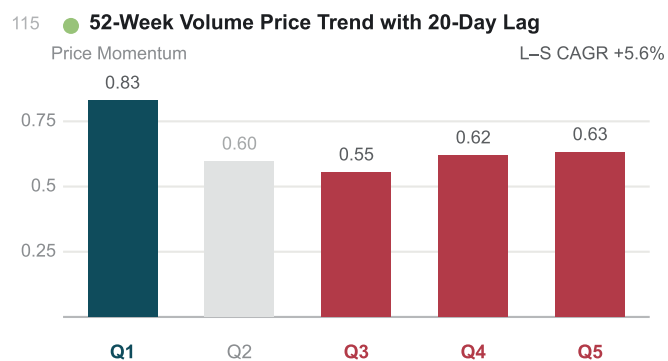


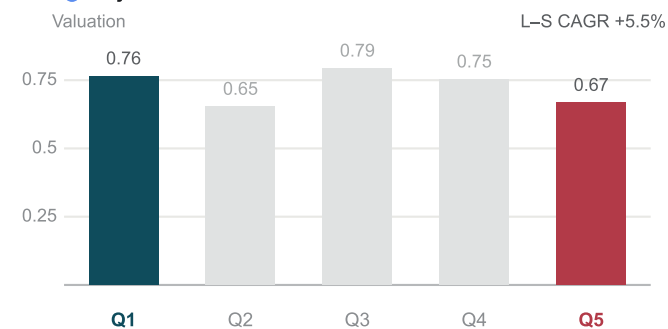
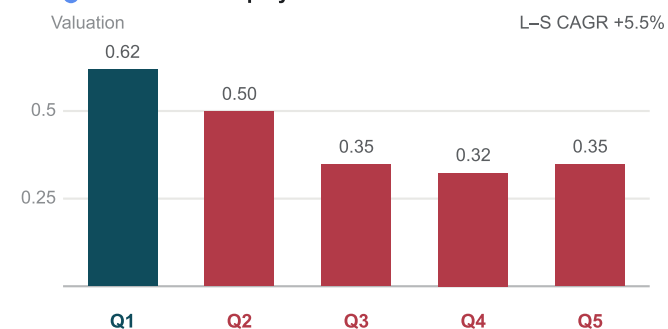
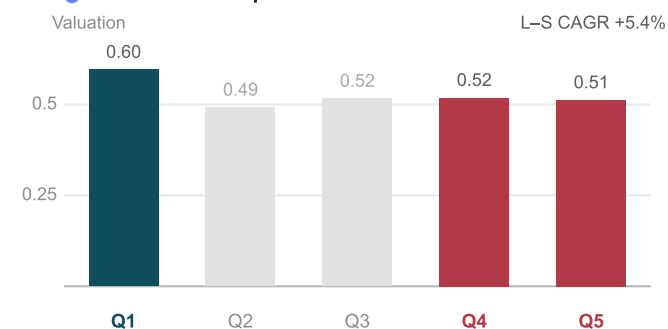
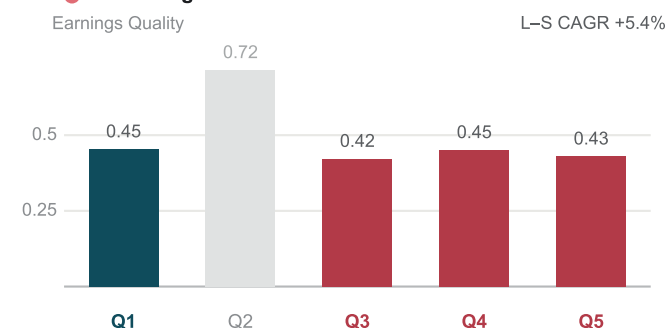
102 ● Cash Ratio

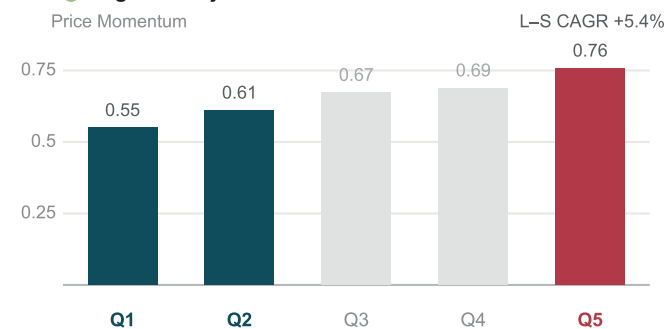
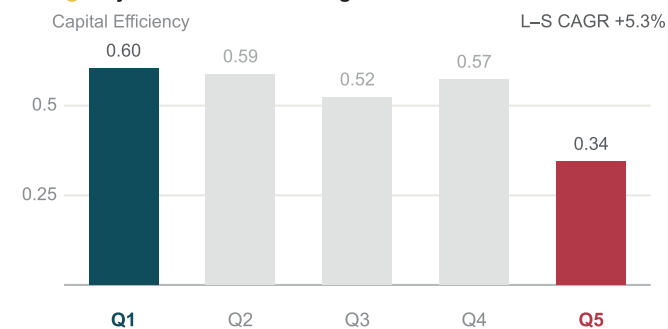
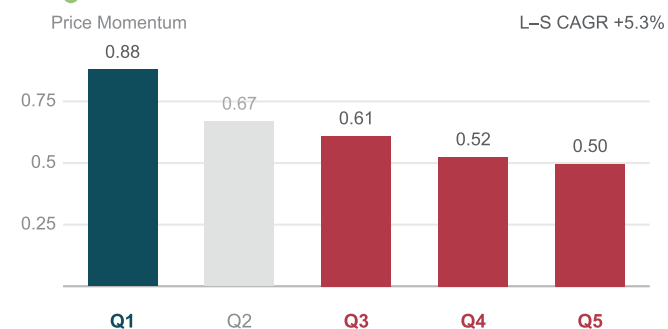
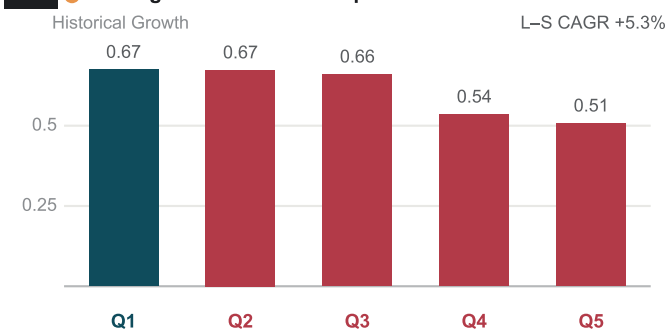
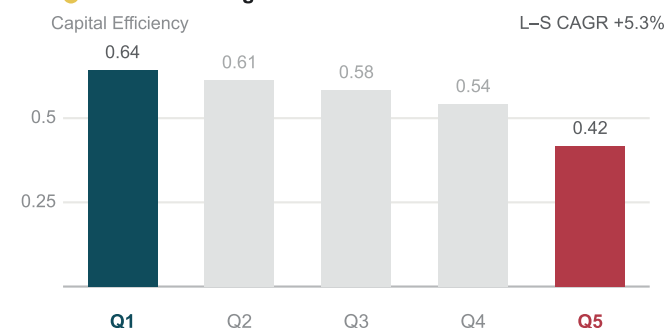
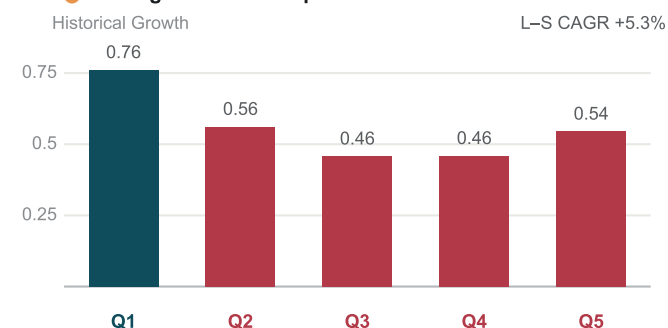


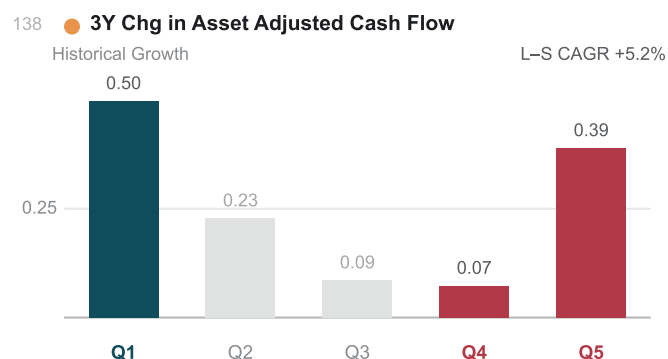
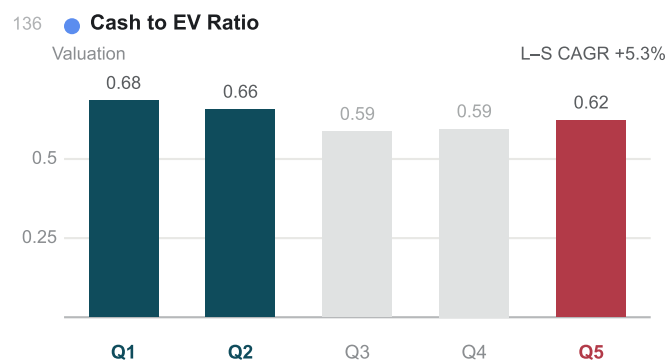
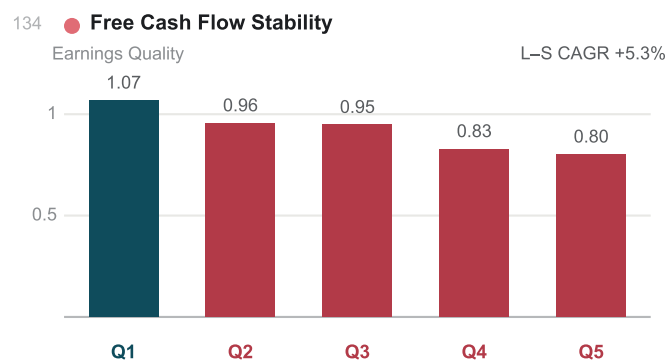
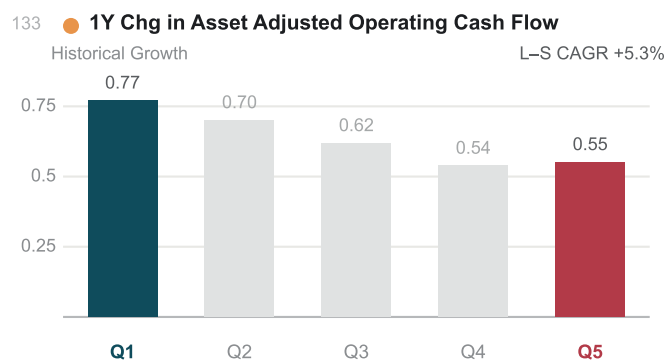


109 ● Dividend Growth**110 ● 1Y Chg in Operating Cash Flow to Price****111 ● 1Y Chg in Free Cash Flow to Price****112 ● 3Y Chg in Asset Adjusted Operating Cash Flow****113 ● Operating Cash Flow Ratio****114 ● 3Y Chg in Free Cash Flow to Price**



121 ● **EBIT to Price**122 ● **Adj Growth Flow to Price**123 ● **Cash Flow to Equity**124 ● **EBITDA to Enterprise Value**125 ● **Adj. EBITDA to Enterprise Value**126 ● **EBIT Margin**

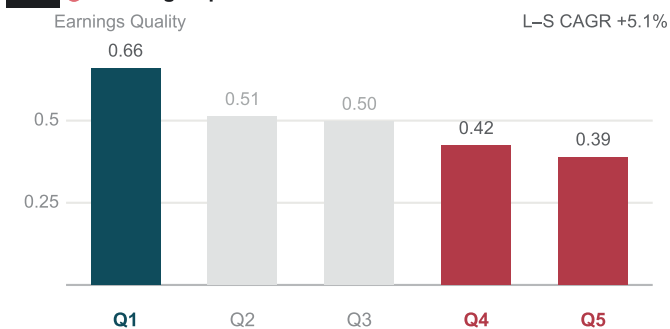
127 ● **Log of Unadjusted Stock Price**128 ● **Adjusted Interest Coverage Ratio**129 ● **15/36 Week Price Ratio**130 ● **1Y Chg in Free Cash Flow per Share**131 ● **Interest Coverage Ratio**132 ● **3Y Chg in Cash Flow per Share**



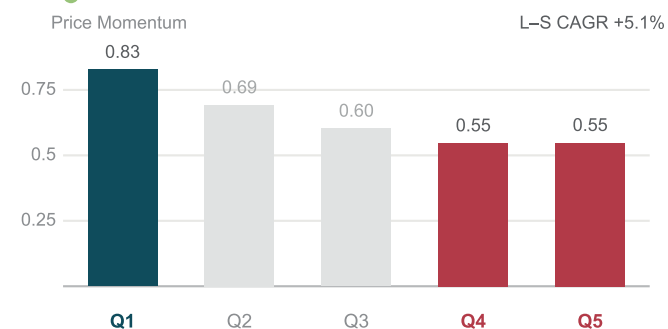
139 ● Adjusted Book to Enterprise Value



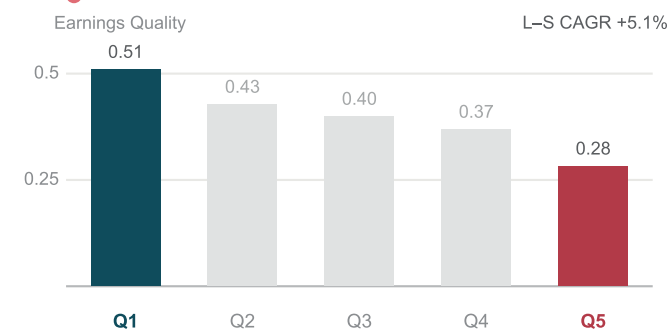
140 ● Working Capital Accruals



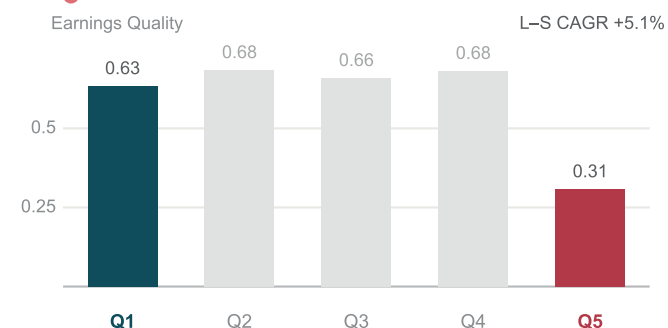
141 ● 9 Month Price Momentum



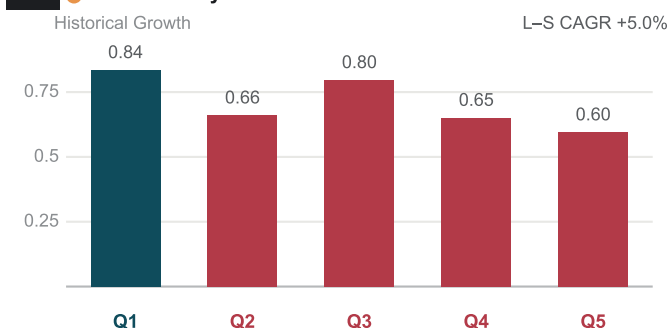
142 ● Free Cash Flow to Sales

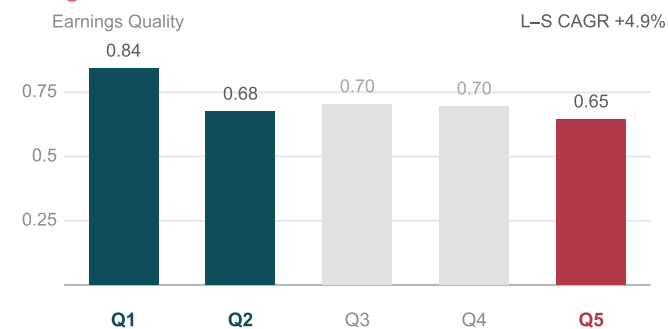
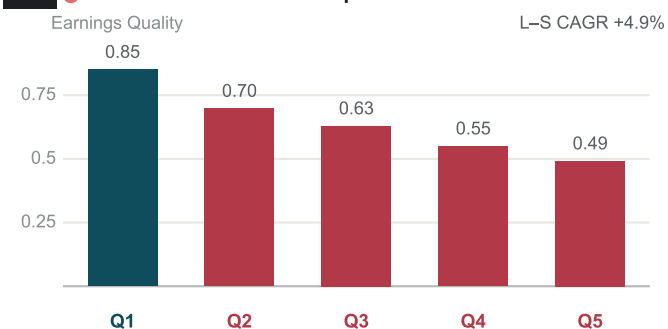
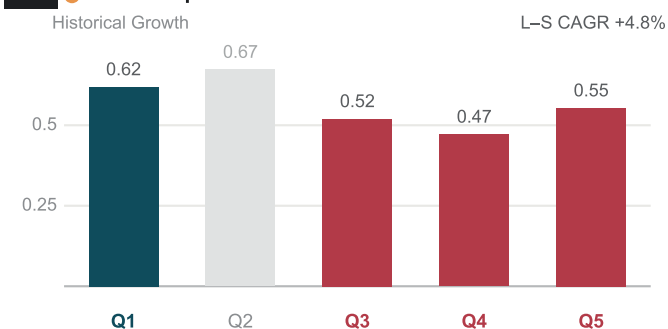
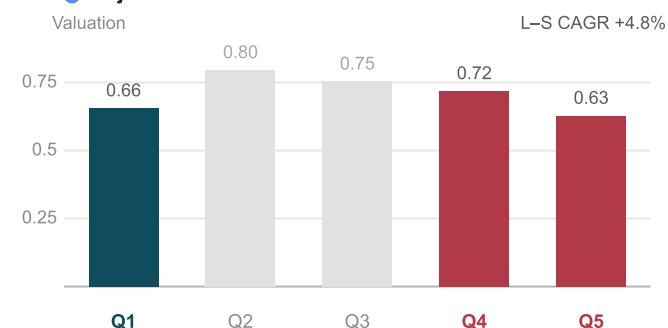


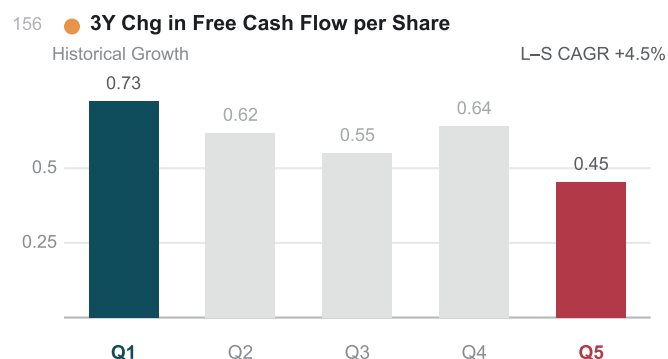
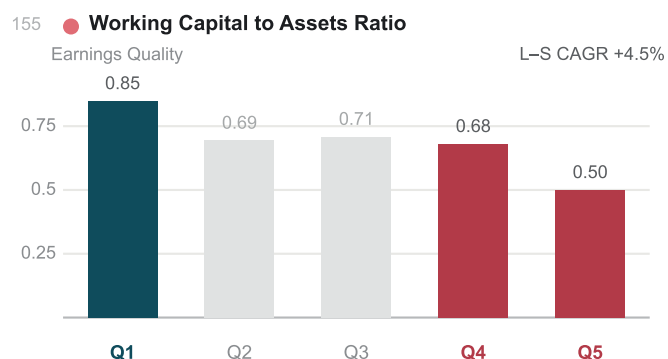
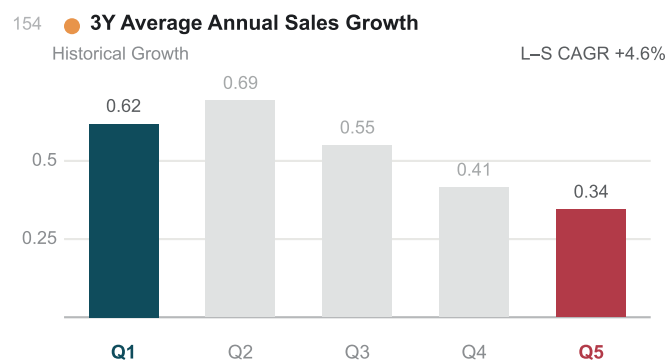
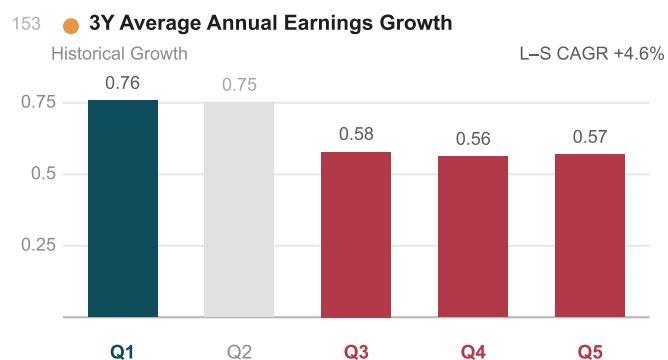
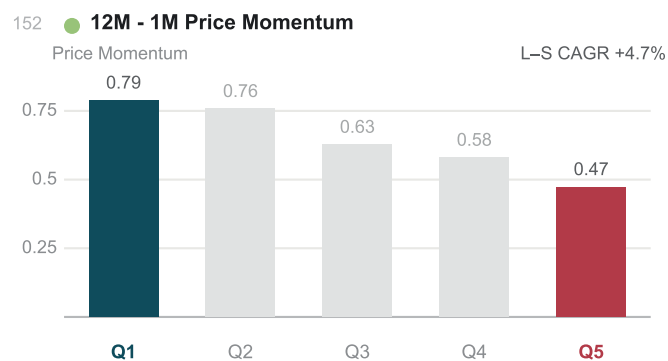
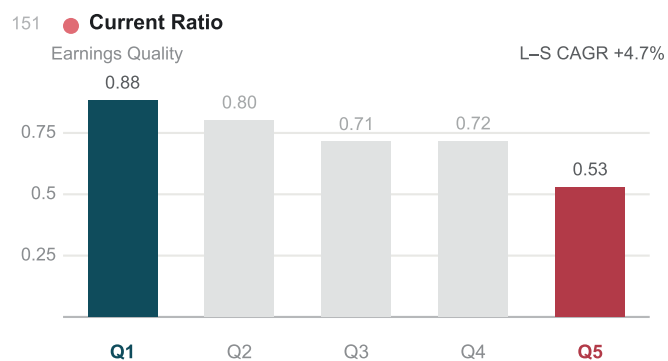
143 ● Asset Turnover

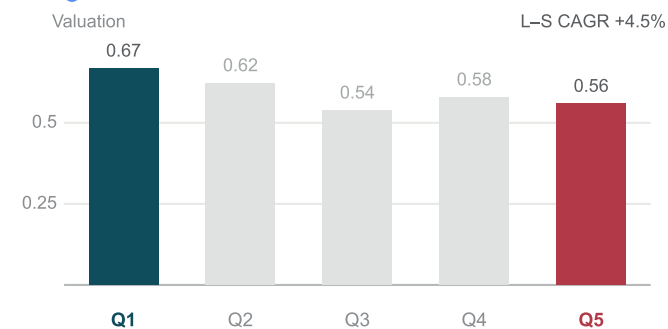
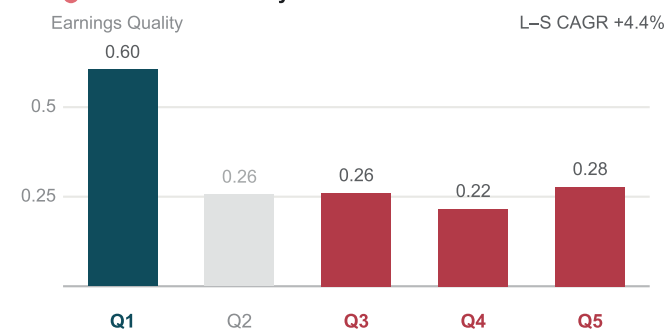
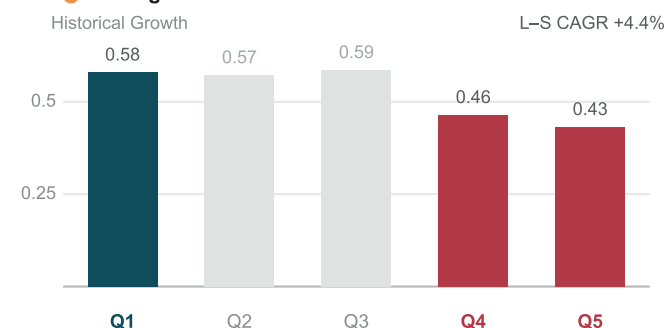
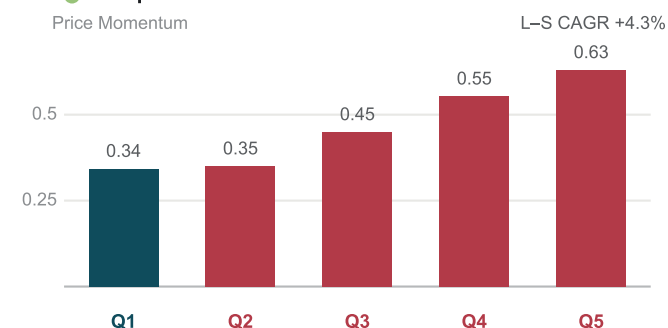


144 ● EPS Stability

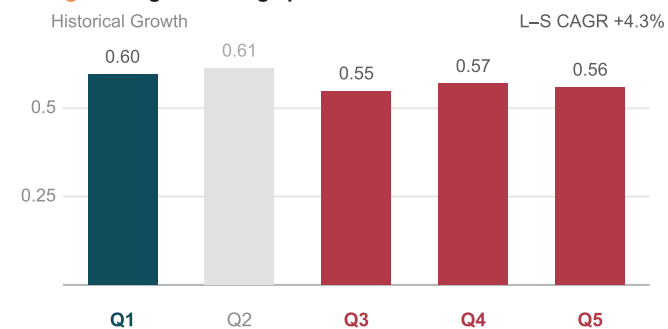


145 ● **6M Chg in Target Price**146 ● **Cash Burn Rate**147 ● **Cash Flow to Invested Capital**148 ● **Dividend per Share Growth**149 ● **1Y Chg in Sales Turnover**150 ● **Adj Forward Free Cash Flow to Price Ratio**

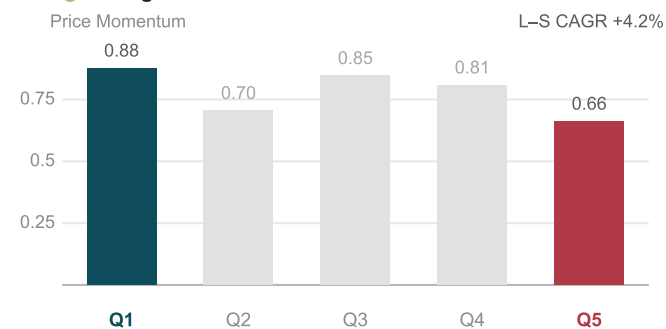


157 ● **1Y Chg in Shares Outstanding**158 ● **Cash to Total Assets**159 ● **Net Income Stability**160 ● **Inventory to Assets Ratio**161 ● **3Y Chg in Sales to Price**162 ● **Sharpe Ratio**

163 ● 1Y Chg in Earnings per Share



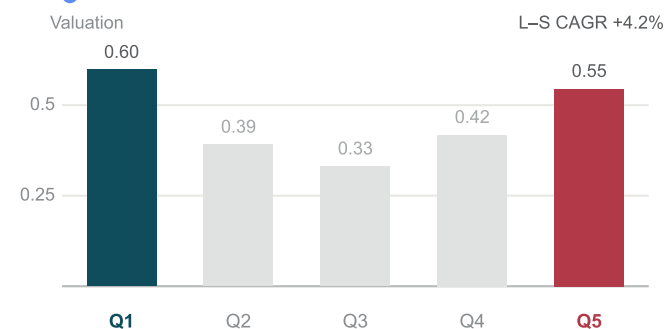
164 ● Change in Number of Holders



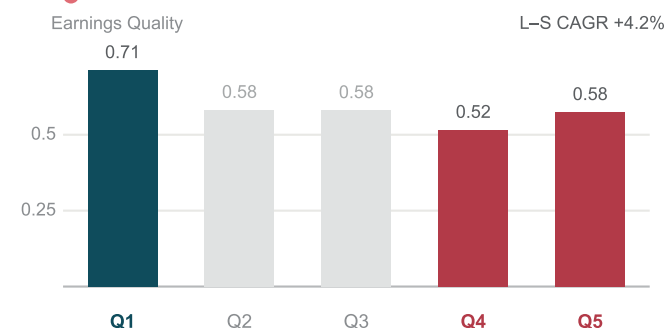
165 ● Return on Equity Stability



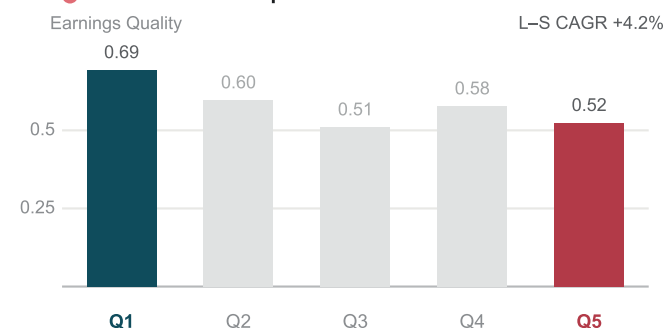
166 ● Net Current Assets to Price Ratio

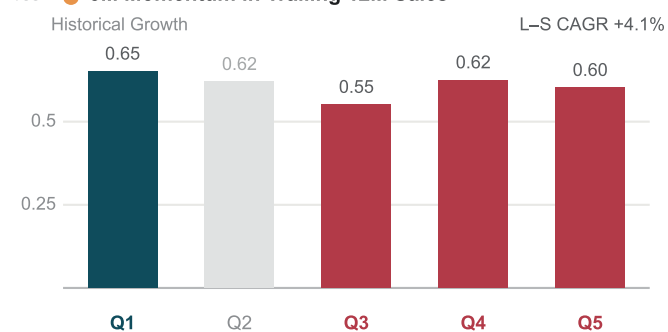
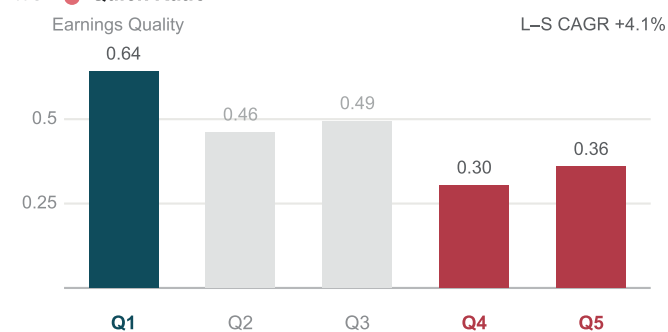
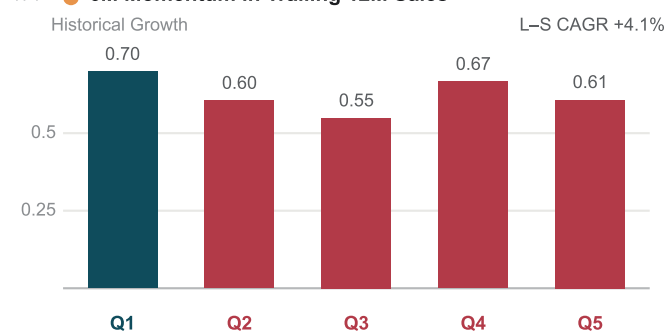
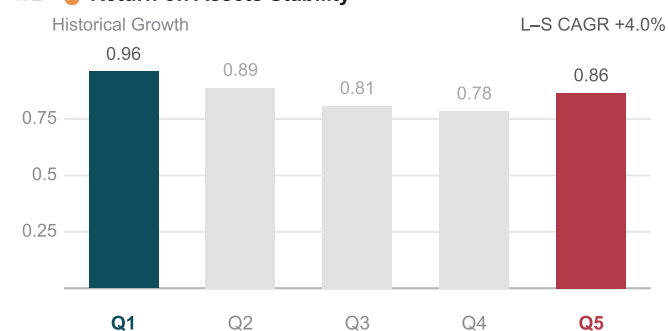
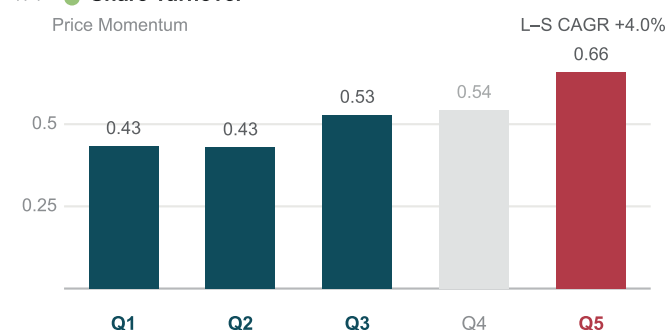


167 ● Accrual Ratio - Cash Flows



168 ● ROA 60 Month Slope



169 ● **3M Momentum in Trailing 12M Sales**170 ● **Quick Ratio**171 ● **6M Momentum in Trailing 12M Sales**172 ● **Return on Assets Stability**173 ● **Number of EPS FY1 Revisions**174 ● **Share Turnover**

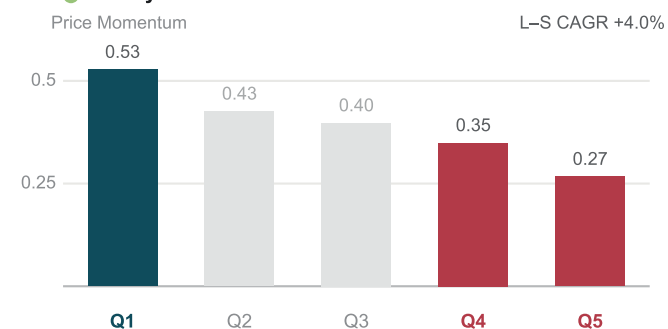
175 ● 3Y Chg in Sales Turnover



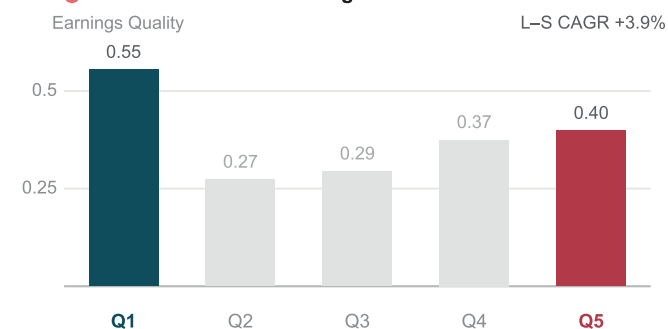
176 ● 26W Relative Price Strength



177 ● 60 Day Coefficient of Variation of Volume to Price



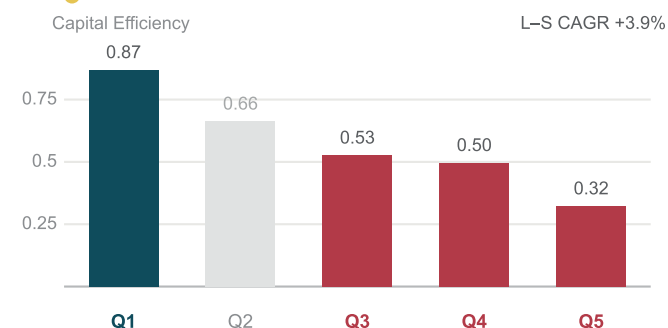
178 ● Sales to Gross Profit Margin Growth

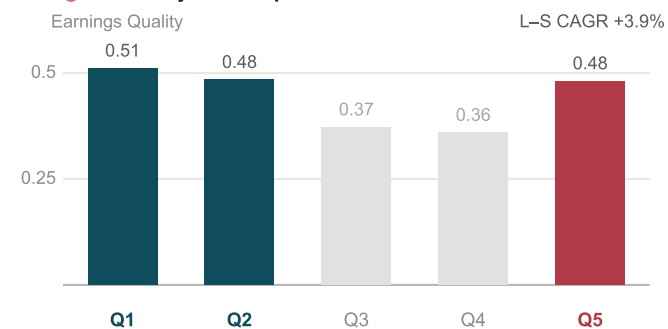
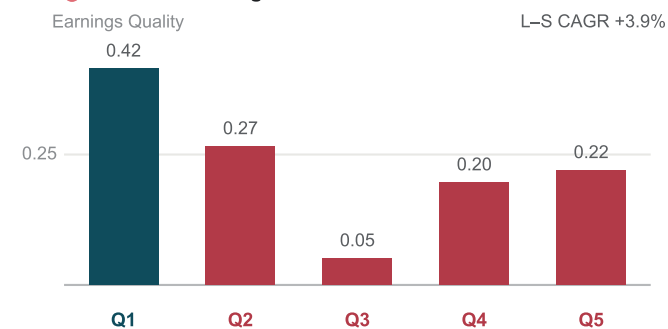
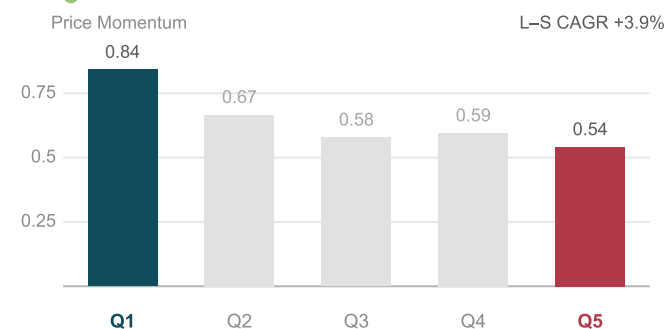
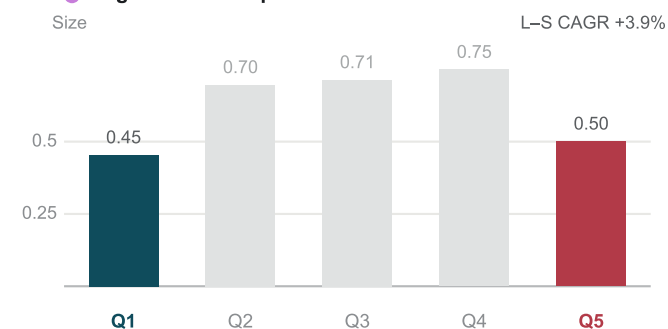


179 ● Cash Flow Return on Invested Capital

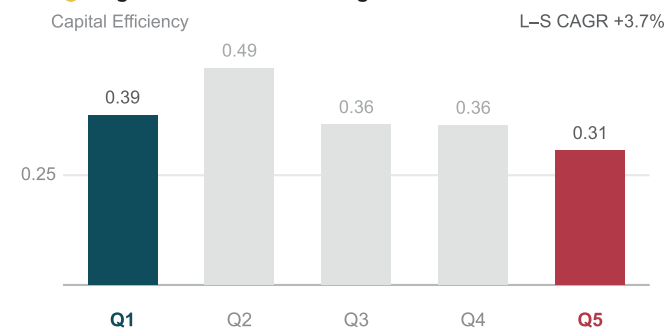


180 ● Gross Profit to Assets

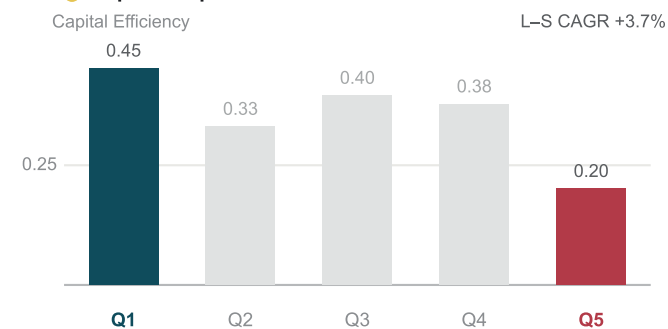


181 ● **Asset Adjusted Capital Investments**182 ● **Retained Earnings to Total Assets**183 ● **4W to 52W Price Oscillator**184 ● **Log of Market Capitalization Cubed**185 ● **50 Day to 200 Day Stock Price**186 ● **3Y Chg in Asset Adjusted Free Cash Flow**

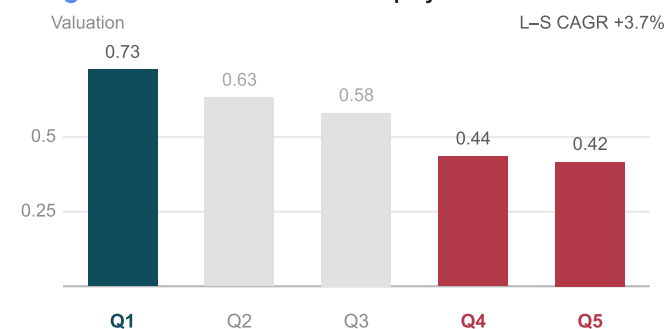
187 ● Degree of Financial Leverage



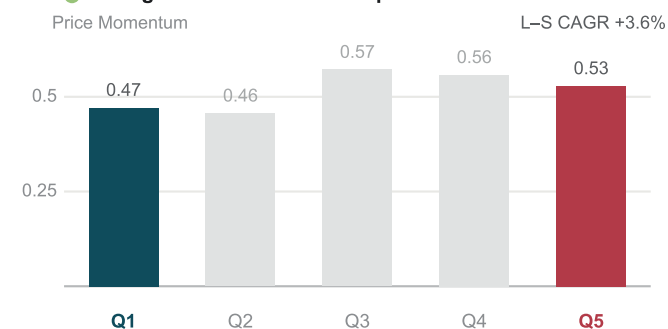
188 ● Capital Acquisition Ratio



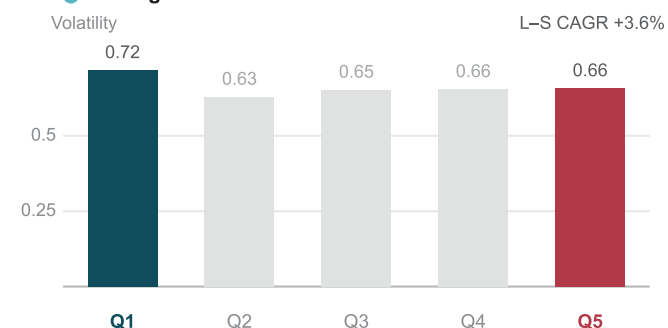
189 ● Free Cash Flow to Common Equity



190 ● Foreign Institution Ownership

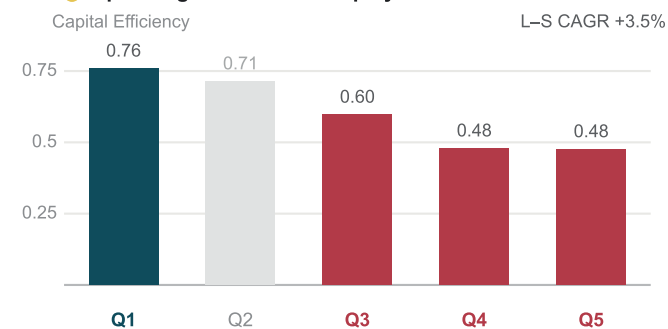
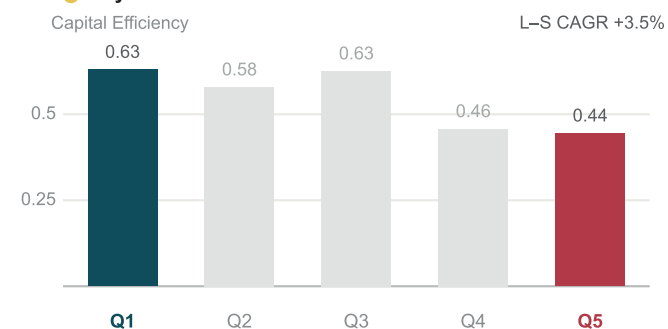
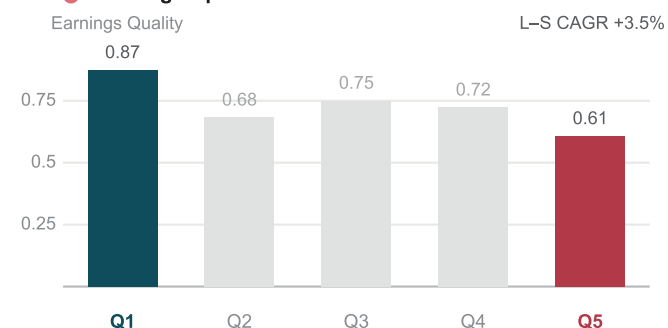


191 ● 1Y Chg in Debt to Assets



192 ● Sustainable Growth Rate

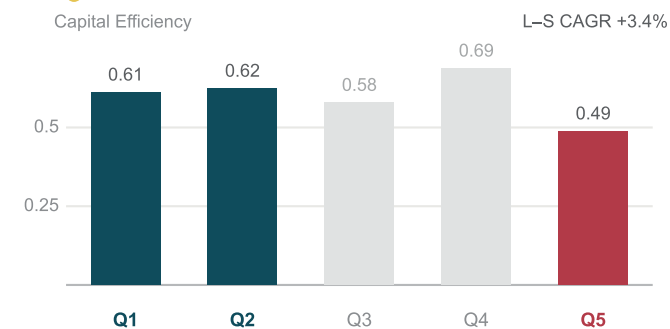


193 ● **Net Profit Margin**194 ● **Operating Cash Flow to Equity**195 ● **Payout Ratio**196 ● **Log Market Cap**197 ● **Working Capital to Total Assets**198 ● **Pretax Income to Net Operating Assets**

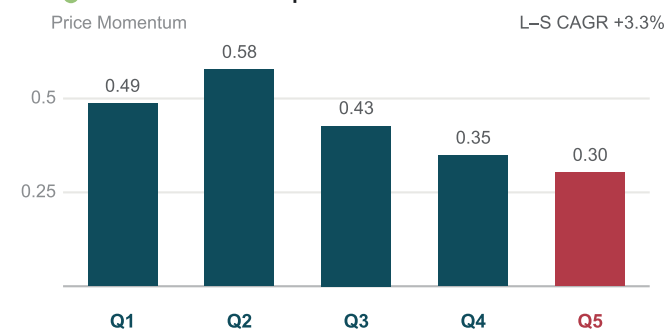
199 ● Risk Adjusted Relative Strength



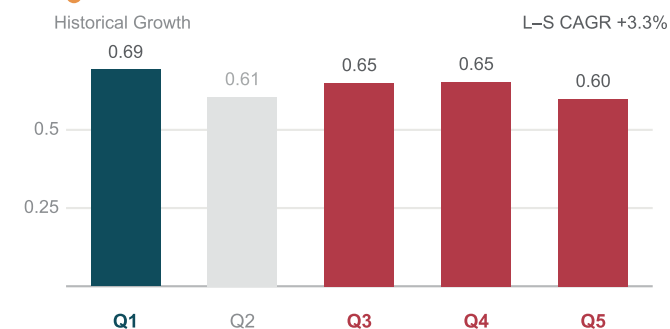
200 ● Asset Growth - 1 Year



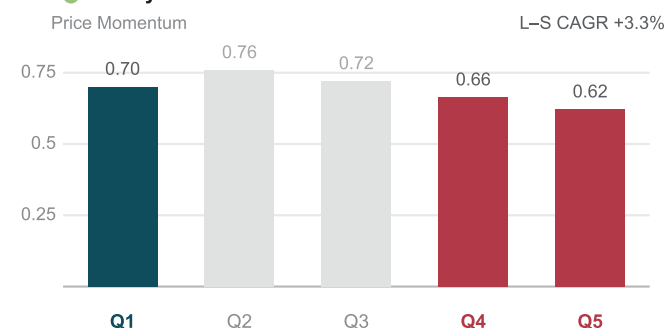
201 ● Institution Ownership Level



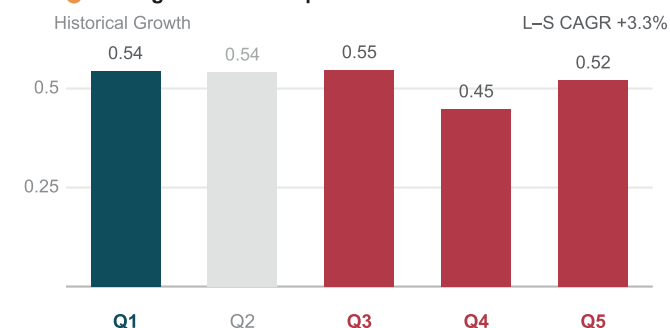
202 ● Num of Quarters of Declines in Receivables and Inventories/...

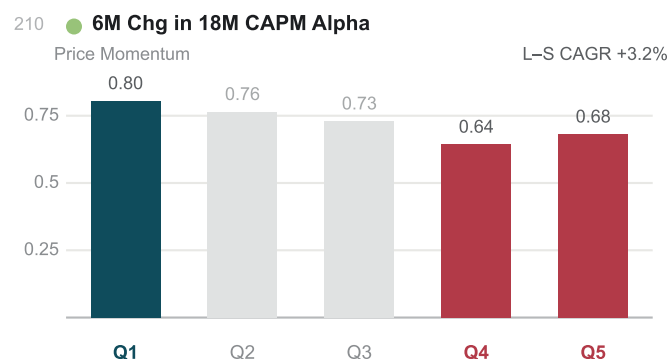
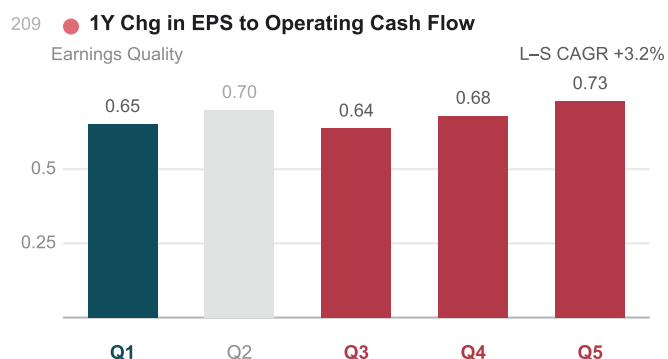
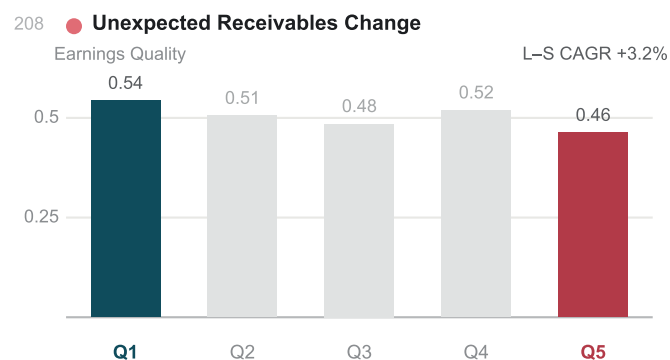
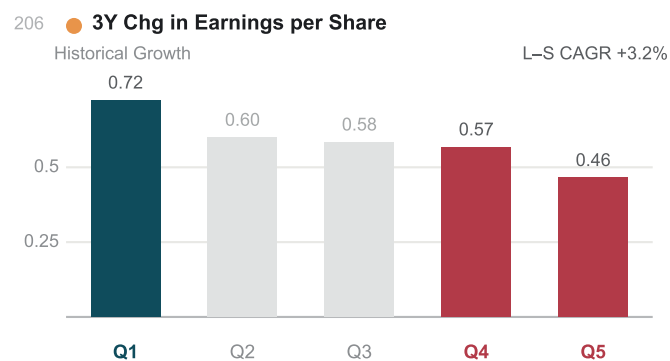
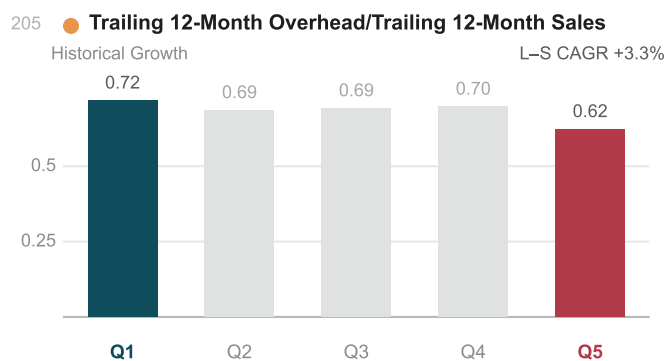


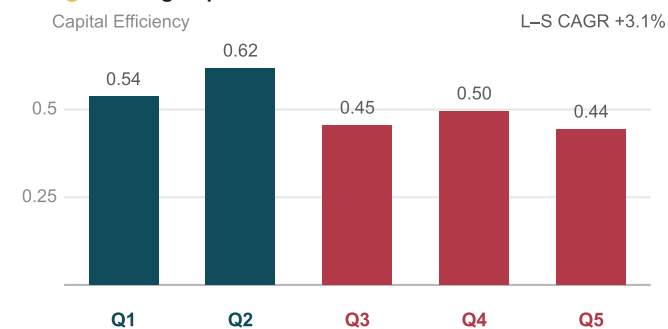
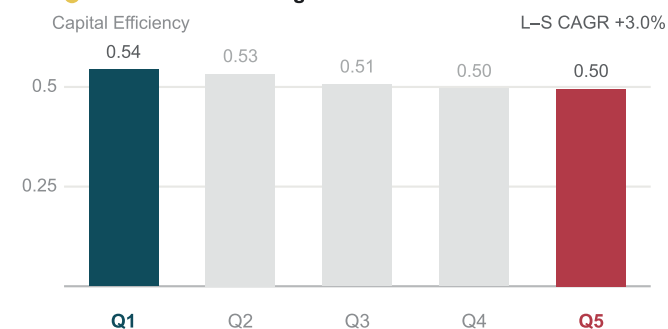
203 ● 10 Day MACD Trend



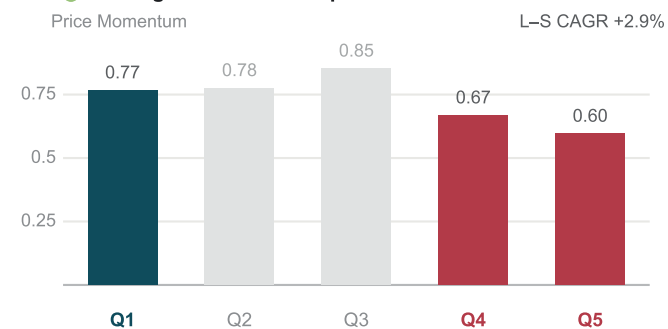
204 ● 1Y Chg in Cash Flow per Share



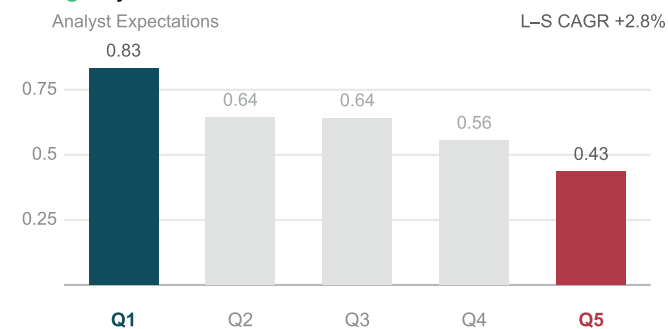


211 ● **Institution Ownership Concentration**212 ● **Working Capital Turnover Ratio**213 ● **Target Price Gap to 6M EMA**214 ● **Year over Year Change of Total Debt**215 ● **Research & Development Intensity**216 ● **Cash Conversion Cycle**

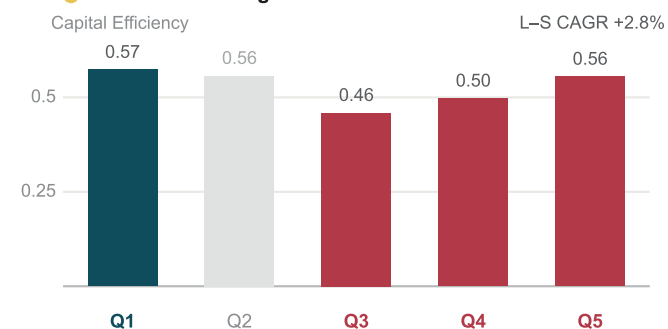
217 ● 6M Chg in 36M CAPM Alpha



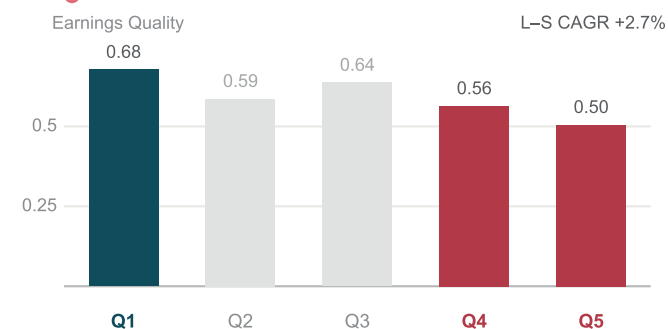
218 ● Adj Number of EPS FY2 Revisions



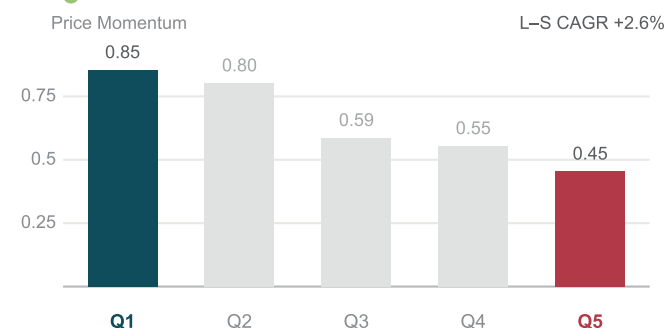
219 ● Pretax Profit Margin



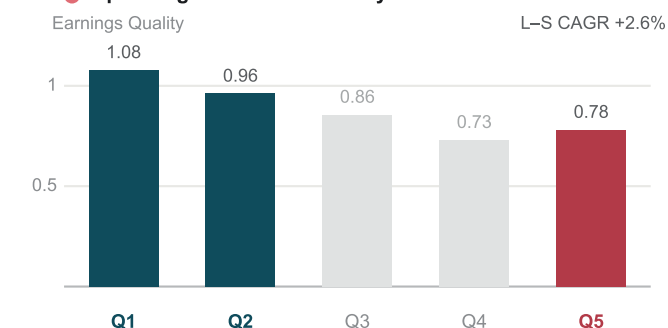
220 ● Accrual Ratio - Balance Sheet



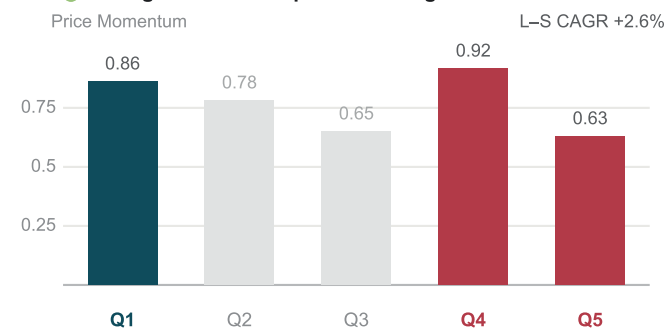
221 ● 180D Price TStat



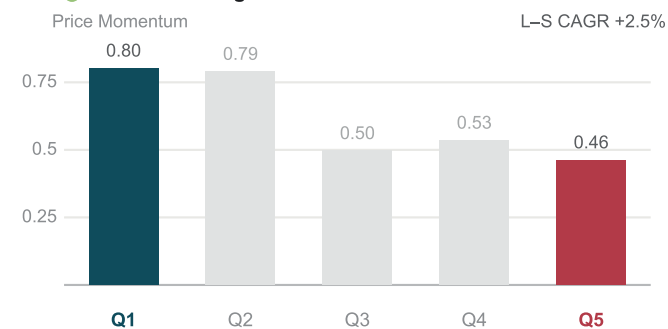
222 ● Operating Cash Flow Stability



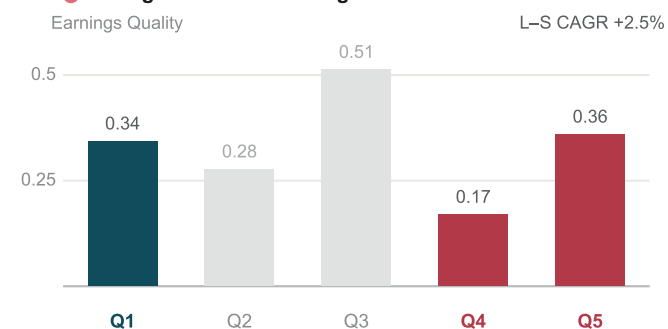
223 ● Change of Ownership Level - Hedge Fund



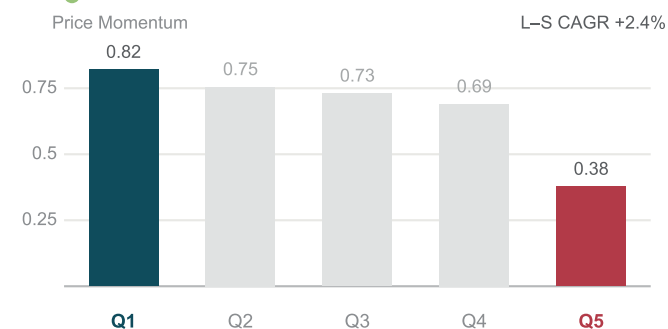
224 ● 6 Month Average Share Turn Over



225 ● Change of Sales to Change of EPS



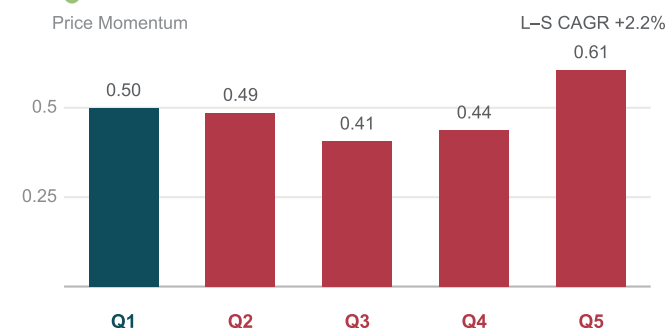
226 ● 26W Price Stat



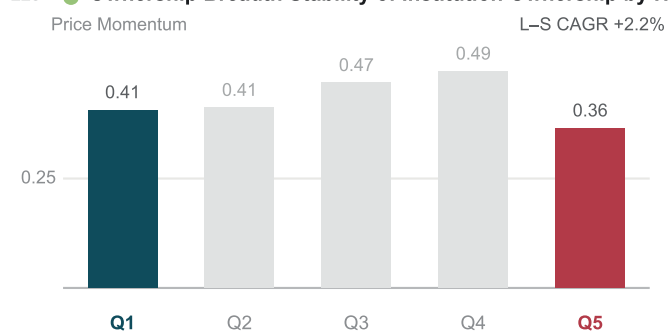
227 ● Ownership Breadth Stability of Institution Ownership by Buyers



228 ● 6 Months Price Reversal



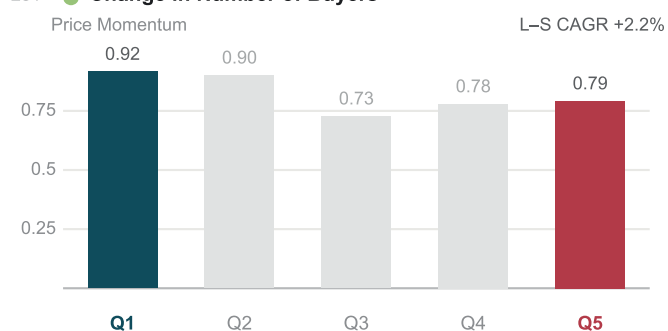
229 ● Ownership Breadth Stability of Institution Ownership by Hol...



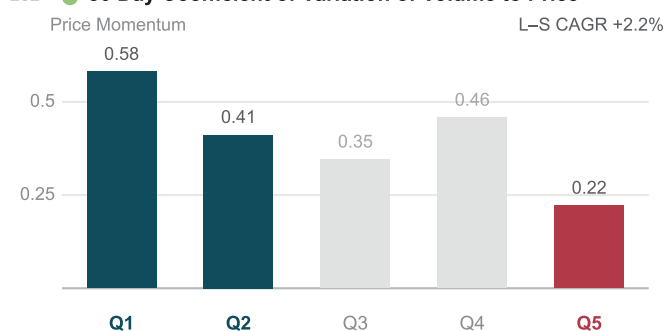
230 ● 6M Chg in Target Price Gap



231 ● Change in Number of Buyers



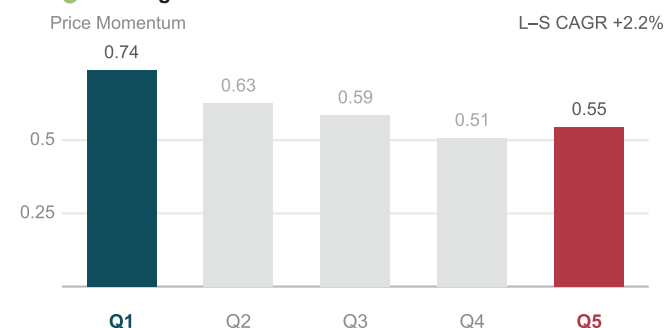
232 ● 30 Day Coefficient of Variation of Volume to Price

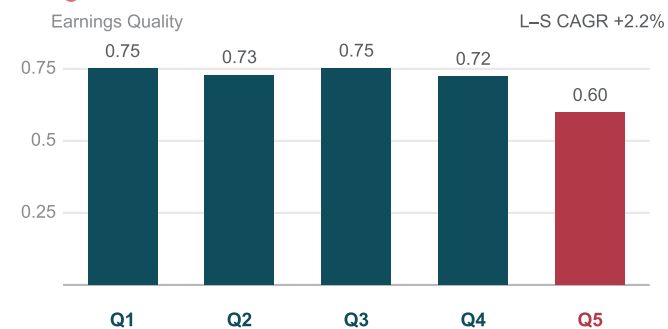
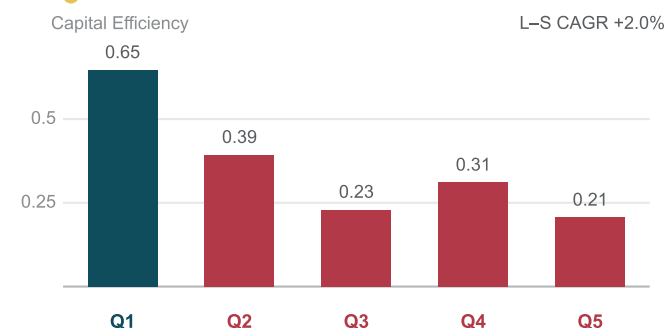


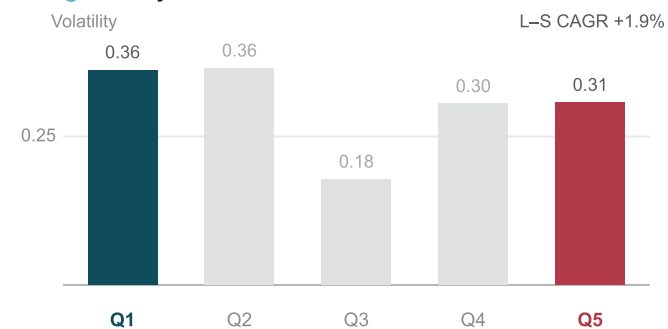
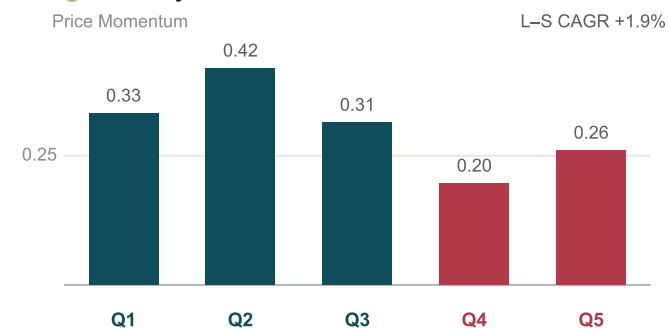
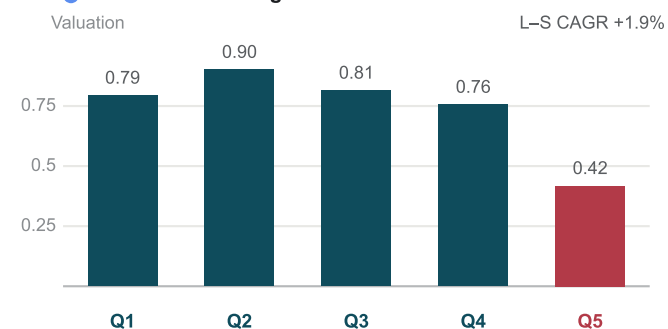
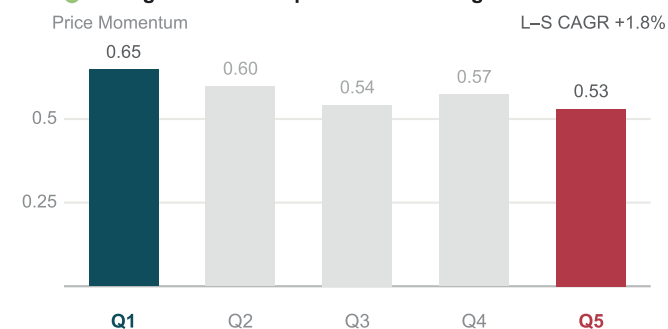
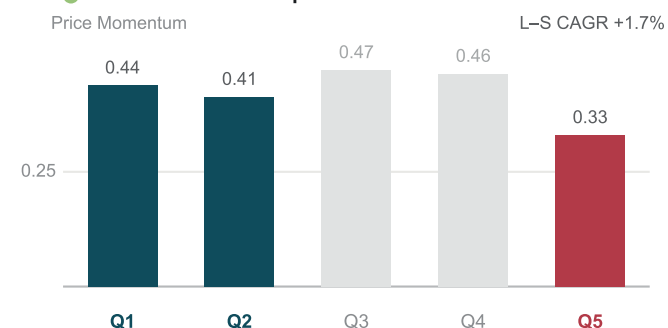
233 ● 1Y Chg in Amihud

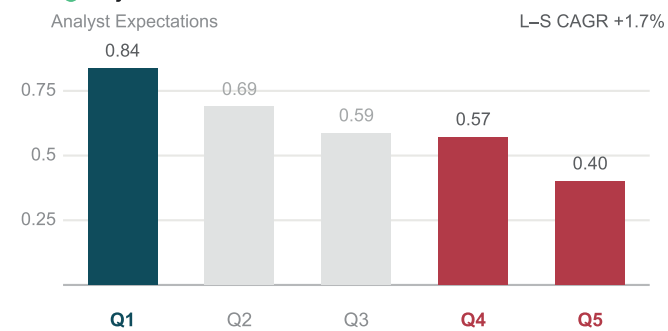
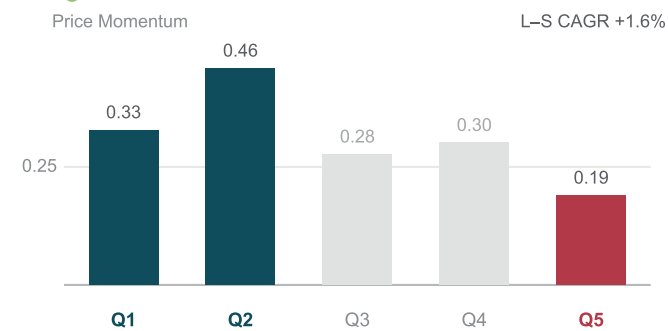
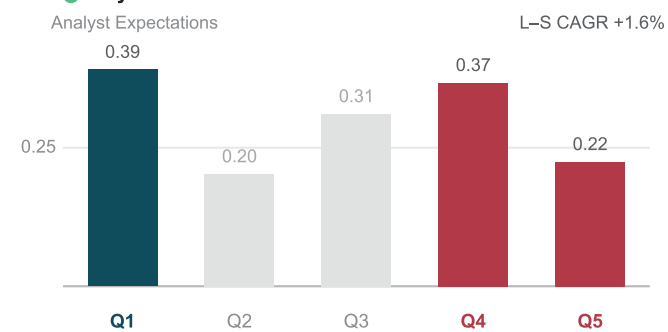
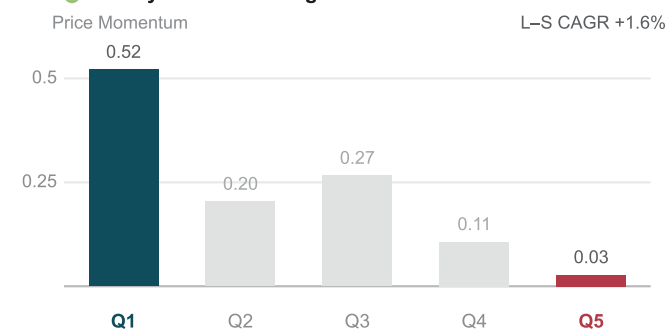


234 ● 52W High Low

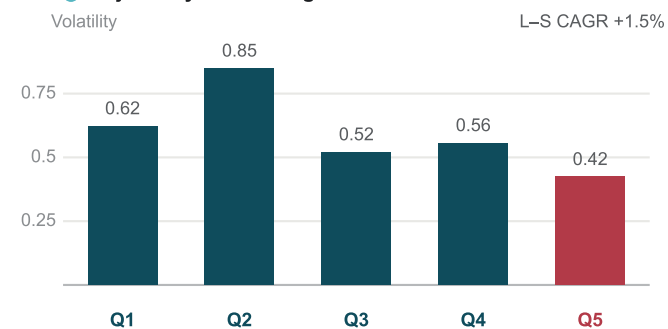


235 ● **Cash to Sales**236 ● **8-Week Change in 12-Month Forward Earnings Consensus Estima...**237 ● **Year over Year Change of Inventory to Assets**238 ● **Volatility adjusted 12M return**239 ● **Normalized ROA**240 ● **Change in Slope of 4-Quarter Trendline through Quarterly Sales**

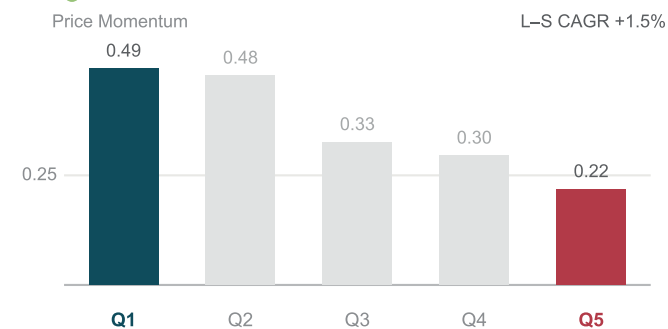
241 ● **130 Day Minimum Return**242 ● **Max Daily Return in the Past 6 Months**243 ● **Return on Net Tangible Assets**244 ● **Change of Ownership Level - All Managers**245 ● **Institution Ownership Turnover**246 ● **6M Chg in 12M CAPM Alpha**

247 ● **Adjusted Number of EPS FY1 Revisions**248 ● **24M Residual Return Variance**249 ● **Buy-to-Sell Recommendation Ratio Less 3M SMA**250 ● **14-Day Relative Strength Index**251 ● **Analyst Earnings Estimate Diffusion**252 ● **3Y Chg in Operating Margin**

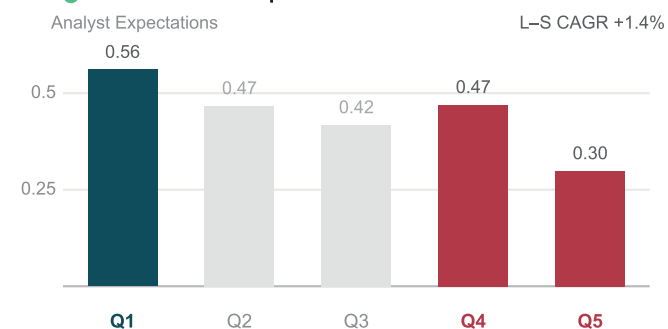
253 ● Adj 50 Day Volume Signal



254 ● Investment Duration



255 ● EBIT Estimate Dispersion



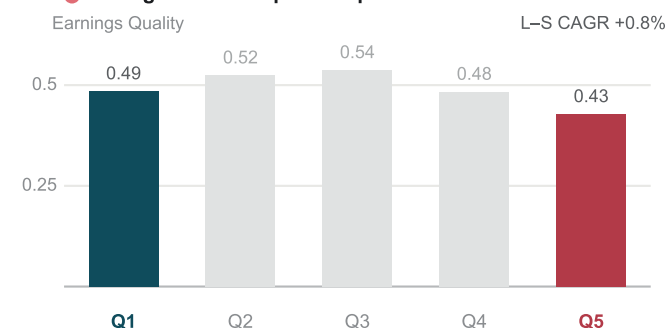
256 ● Sales Acceleration



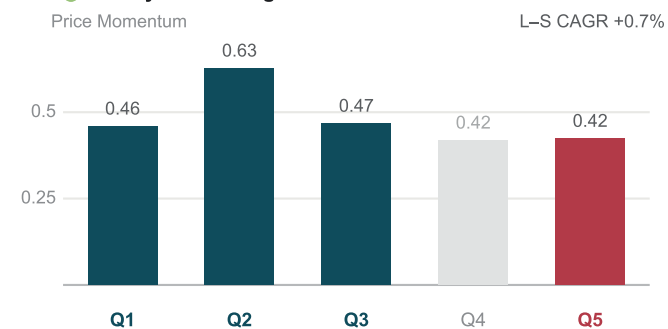
257 ● Revenue Estimate Dispersion



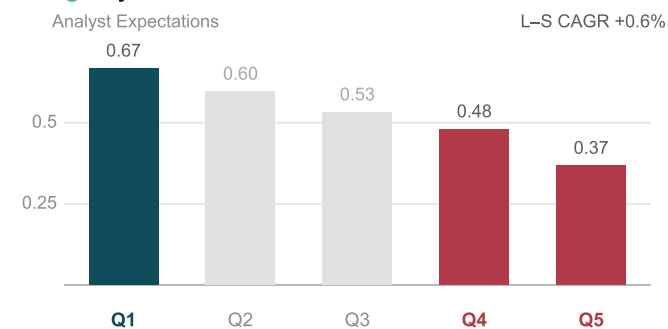
258 ● Change in TTM Depr. to Capex



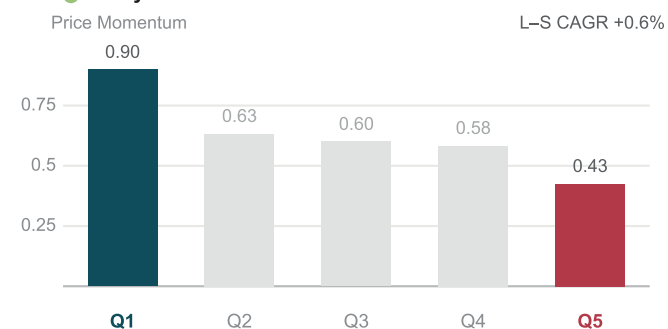
259 ● 50 Day Volume Signal



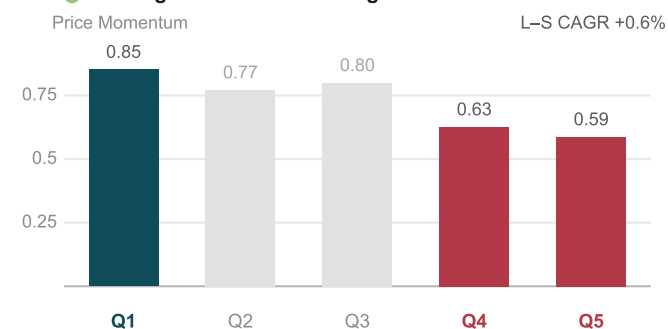
260 ● Buy-to-Sell Recommendation Ratio Less 3M EMA



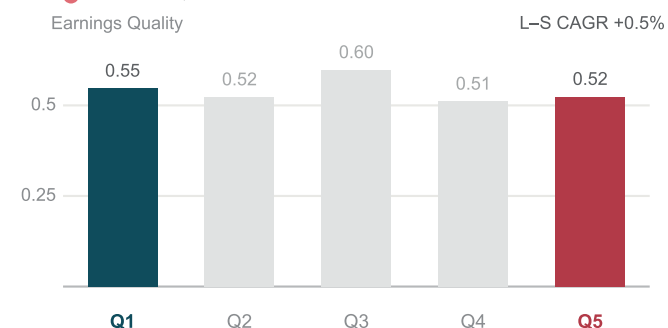
261 ● 5 Day Price Reversal



262 ● Closing Price to 52 Week High



263 ● ROE 20 Qtr Standard Deviation



264 ● 1Y Chg in Share Turnover

